

ZUSTANDSBERICHT DER DATENTRÄGER · SUPERMICRO X13SAE-F

SMART-Bericht aller 16 Datenträger

Vollständige Auswertung mit ungekürzten Rohausgaben im Anhang, damit sich jede Angabe nachprüfen lässt. Zwei Auffälligkeiten sind offengelegt.

Erhoben 27./28. August 2026 Werkzeug smartctl 7.4 Umfang 10 SAS-SSD · 4 HDD · 2 SATA-SSD

Das Wichtigste zuerst

DATENTRÄGER	STATUS NICHT OK	UMGELAGERTE SEKTOREN	SCHWEBENDE SEKTOREN	UNLESBARE SEKTOREN	SSD-ABNUTZUNG
16	0	0	0	0	0 %

Alle sechzehn Datenträger melden einen fehlerfreien Gesundheitsstatus, und auf keinem einzigen steht ein Zähler für Medienschäden über null. Auf jedem lief vor der Erhebung ein kurzer Selbsttest, alle ohne Beanstandung.

BEMERKENSWERT

Die zehn SAS-SSDs haben zwischen **143 und 375 Betriebsstunden** bei einem Abnutzungsgrad von **0 %**. Alle stammen aus derselben Fertigungswoche (KW 46/2021) und sind praktisch ungenutzt.

OFFENGELEGT

Zwei der vier 10-TB-Platten zeigen Zähler, die nicht auf null stehen: eine hat **Command Timeouts**, eine **drei CRC-Fehler** auf der Schnittstelle. Beides betrifft die Anbindung, nicht das Medium — die Einzelheiten stehen unten. Auf beiden Platten sind alle Zähler für Medienschäden null.

ERWEITERTE SELBSTTESTS BESTANDEN

Auf allen vier 10-TB-Platten lief am 27./28.08. ein **erweiterter Selbsttest**, der die vollständige Oberfläche liest — rund 15 Stunden je Platte. Alle vier melden `Completed without error`. Danach standen umgelagerte, schwebende und unlesbare Sektoren unverändert auf null, und auch die beiden auffälligen Zähler sind nicht weitergelaufen.

10 × SAS-SSD 3,2 TB

HPE VO003200JWZJR, Firmware HPD5, alle Fertigungswoche 46/2021

SERIENNUMMER	STUNDEN	STARTZYKLEN	TEMP.	ABNUTZUNG	LESEFEHLER	SCHREIBFEHLER	STATUS
S5KUNA0RB02927	143	66	27 °C	0 %	0	0	OK
S5KUNA0RB02884	145	73	28 °C	0 %	0	0	OK
S5KUNA0RB02923	196	48	34 °C	0 %	0	0	OK
S5KUNA0RB02929	196	48	35 °C	0 %	0	0	OK
S5KUNA0RB02935	196	48	34 °C	0 %	0	0	OK
S5KUNA0RB02886	198	47	34 °C	0 %	0	0	OK
S5KUNA0RB02911	303	71	34 °C	0 %	0	0	OK
S5KUNA0RB02930	314	76	35 °C	0 %	0	0	OK
S5KUNA0RB03018	364	72	34 °C	0 %	0	0	OK
S5KUNA0RB03024	375	72	36 °C	0 %	0	0	OK

Der Wechselrahmen fasst acht Laufwerke. Für die Messung der beiden zusätzlichen wurden vorübergehend zwei eingebaute entnommen — deshalb sind alle zehn erfasst, waren aber nie zehn gleichzeitig im Gerät.

ZUM ZÄHLER „NICHT-MEDIENBEZOGENE FEHLER“

In den Rohausgaben steht je Laufwerk ein Wert zwischen 1 714 und 4 288. Dieser Zähler erfasst Ereignisse auf dem Bus — Zurücksetzungen, Aushandlungen, Verbindungsereignisse — und **keine Datenfehler**. Bei SAS-Laufwerken in Wechselrahmen sind Werte dieser Größenordnung üblich und für sich genommen unauffällig. Die maßgeblichen Zähler für Datenintegrität stehen bei allen zehn auf null.

Die Betriebsstunden stehen bei SAS-Laufwerken nicht als eigenes Attribut zur Verfügung. Die Werte oben stammen aus der Spalte `LifeTime` des Selbsttestprotokolls, die den Stundenstand zum Zeitpunkt des Tests festhält — der Selbsttest vom 27.08. hat sie also überhaupt erst sichtbar gemacht.

4 × Festplatte 10 TB

Seagate ST10000NM0086, Baureihe Exos

SERIENNUMMER	STUNDEN	TEMP.	UMGELAGERT	SCHWEBEND	UNLESBAR	CRC	TIMEOUTS	OBERFLÄCHENTEST	STATUS
ZA2DN4VV	4 363	38 °C	0	0	0	0	0	bestanden	PASSED
ZA2DXWL9	8 622	36 °C	0	0	0	0	122	bestanden	PASSED
ZA2ATQGT	8 888	40 °C	0	0	0	3	0	bestanden	PASSED
ZA2A4DQ1	37 805	40 °C	0	0	0	0	0	bestanden	PASSED

ZA2DXWL9 — Command Timeouts, Rohwert 122

Der schlechteste normalisierte Wert dieses Zählers ist auf 42 gefallen (Neuwert 100, Warnschwelle 0). Das heißt: Die Platte hat irgendwann in ihrer Laufzeit auf Befehle nicht rechtzeitig geantwortet. Übliche Ursachen sind Kabel, Backplane oder Controller, seltener die Platte selbst. **Alle Zähler für Medienschäden stehen auf null**, umgelagerte und schwebende Sektoren ebenso.

Der erweiterte Selbsttest vom 28.08. ist fehlerfrei durchgelaufen und hat dabei die vollständige Oberfläche gelesen. Der Timeout-Zähler ist seither nicht weitergelaufen. Ein Medienschaden ist damit ausgeschlossen.

ZA2ATQGT — drei CRC-Fehler auf der Schnittstelle

Übertragungsfehler zwischen Platte und Controller, praktisch immer eine Frage von Kabel oder Kontakt und kein Hinweis auf Plattenschäden. Der Zähler zählt nicht zurück; drei Ereignisse über 8 888 Stunden sind unkritisch. Auch hier alle Medienzähler auf null.

Auch diese Platte hat den erweiterten Selbsttest fehlerfrei bestanden, und der CRC-Zähler steht unverändert bei drei.

ZA2A4DQ1 — die älteste der vier

37 785 Stunden sind gut vier Jahre Dauerlauf. Die Platte ist fehlerfrei, aber sie ist diejenige, die man zuerst im Blick behalten sollte.

ZU DEN ROHWERTEN IM ANHANG

Bei Seagate stehen bei `Raw_Read_Error_Rate` und `Seek_Error_Rate` Rohwerte in Millionenhöhe. **Das sind keine Fehler**. Seagate kodiert dort Zähler für gelesene beziehungsweise gesuchte Sektoren mit; der Rohwert von `Raw_Read_Error_Rate` ist bei allen vier Platten identisch mit dem von `Hardware_ECC_Recovered`, also der Zahl regulär korrigierter Lesevorgänge. Maßgeblich ist der normalisierte Wert: Er liegt hier zwischen 72 und 90 bei einer Warnschwelle von 44 — also mit deutlichem Abstand im unauffälligen Bereich.

2 × SATA-SSD 800 GB

HPE VK0800GDJYA — bilden gespiegelt den Systemverbund

SERIENNUMMER	STUNDEN	TEMP.	UMGELAGERT	SCHWEBEND	UNLESBAR	CRC	STATUS
BTWL341407UP800RGN	92 498	26 °C	0	0	0	0	PASSED
BTWL3414088B800RGN	92 489	26 °C	0	0	0	0	PASSED

ZUR EINORDNUNG

92 500 Stunden sind rund **zehneinhalb Jahre Dauerlauf**. Beide sind vollständig fehlerfrei und melden `PASSED`, aber das ist ein hohes Alter. Wer den Systemverbund langfristig darauf laufen lassen will, sollte Ersatz einplanen. Die Spiegelung fängt den Ausfall einer der beiden ab.

Wie erhoben wurde

```
smartctl -t short /dev/sdX      # kurzer Selbsttest, vorab auf allen 16  
smartctl -t long  /dev/sdX      # erweiterter Test, auf den vier 10-TB-Platten  
smartctl -x       /dev/sdX      # vollstaendige Ausgabe  
smartctl -l selftest /dev/sdX    # Selbsttestprotokoll
```

Die Berichte sind nach Seriennummer benannt, nicht nach Gerätenamen: Die Zuordnung `/dev/sdX` ändert sich zwischen Neustarts und taugt nicht zur Identifikation. Der Anhang enthält die ungekürzte Ausgabe je Datenträger.

Anhang: vollständige Rohausgaben

Ungekürzt, ein Datenträger je Seite. Jede Angabe der Zusammenfassung lässt sich hier nachschlagen.

S5KUNA0RB02927

```
#####
# Datentraeger /dev/sdf Seriennummer S5KUNA0RB02927
# Erhoben am 2026-08-28 19:20:56 CEST auf Supermicro X13SAE-F
#####

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
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=== START OF INFORMATION SECTION ===
Vendor:                HPE
Product:               V0003200JWZJR
Revision:              HPD5
Compliance:            SPC-5
User Capacity:         3,200,631,791,616 bytes [3.20 TB]
Logical block size:    512 bytes
Physical block size:   4096 bytes
LU is resource provisioned, LBPRZ=1
Rotation Rate:         Solid State Device
Form Factor:           2.5 inches
Logical Unit id:        0x50000f0b01b653e0
Serial number:         S5KUNA0RB02927
Device type:           disk
Transport protocol:    SAS (SPL-4)
Local Time is:         Fri Aug 28 19:20:56 2026 CEST
SMART support is:      Available - device has SMART capability.
SMART support is:      Enabled
Temperature Warning:   Enabled
Read Cache is:         Enabled
Writeback Cache is:    Disabled

=== START OF READ SMART DATA SECTION ===
SMART Health Status: OK

Percentage used endurance indicator: 0%
Current temperature = 35
Lifetime maximum temperature = 56
Lifetime minimum temperature = 7
Maximum temperature since power on = 35
Minimum temperature since power on = 16
Manufactured in week 46 of year 2021
Accumulated start-stop cycles: 66
Specified load-unload count over device lifetime: 0
Accumulated load-unload cycles: 0
Grown defect list length=1 bytes [unknown number of elements]

Error counter log:
      Errors Corrected by      Total   Correction   Gigabytes   Total
      ECC      by      rereads/   errors   algorithm   processed   uncorrected
      fast | delayed  rewrites  corrected  invocations [10^9 bytes]  errors
read:          0         0          0         0         0         0.000         0
write:         0         0          0         0         0         0.000         0

Non-medium error count:           1728

Pending defect count:0 Pending Defects
SMART Self-test log
Num  Test              Status              segment  LifeTime  LBA_first_err [SK ASC ASQ]
     Description
# 1  Background short  Completed          -         143             - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]

Background scan results log
Status: scan is active
Accumulated power on time, hours:minutes 163:29 [9809 minutes]
Number of background scans performed: 0, scan progress: 3.17%
Number of background medium scans performed: 0
Device does not support General statistics and performance logging

Protocol Specific port log page for SAS SSP
relative target port id = 1
```

```
generation code = 2
number of phys = 1
phy identifier = 0
  attached device type: SAS or SATA device
  attached reason: loss of dword synchronization
  reason: loss of dword synchronization
  negotiated logical link rate: phy enabled; 12 Gbps
  attached initiator port: ssp=1 stp=1 smp=1
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b653e2
  attached SAS address = 0x500304802485ab03
  attached phy identifier = 2
  Invalid DWORD count = 0
  Running disparity error count = 0
  Loss of DWORD synchronization count = 0
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0
relative target port id = 2
generation code = 2
number of phys = 1
phy identifier = 1
  attached device type: no device attached
  attached reason: unknown
  reason: unknown
  negotiated logical link rate: phy enabled; unknown
  attached initiator port: ssp=0 stp=0 smp=0
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b653e3
  attached SAS address = 0x0
  attached phy identifier = 0
  Invalid DWORD count = 0
  Running disparity error count = 0
  Loss of DWORD synchronization count = 0
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0

===== Selbsttestprotokoll =====
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=== START OF READ SMART DATA SECTION ===
SMART Self-test log
Num Test Status segment LifeTime LBA_first_err [SK ASC ASQ]
Description number (hours)
# 1 Background short Completed - 143 - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]
```

S5KUNA0RB02884

```
#####
# Datentraeger /dev/sda Seriennummer S5KUNA0RB02884
# Erhoben am 2026-08-28 19:20:50 CEST auf Supermicro X13SAE-F
#####

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
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=== START OF INFORMATION SECTION ===
Vendor:                HPE
Product:               V0003200JWZJR
Revision:              HPD5
Compliance:            SPC-5
User Capacity:         3,200,631,791,616 bytes [3.20 TB]
Logical block size:    512 bytes
Physical block size:   4096 bytes
LU is resource provisioned, LBPRZ=1
Rotation Rate:         Solid State Device
Form Factor:           2.5 inches
Logical Unit id:       0x50000f0b01b65130
Serial number:         S5KUNA0RB02884
Device type:           disk
Transport protocol:    SAS (SPL-4)
Local Time is:         Fri Aug 28 19:20:50 2026 CEST
SMART support is:      Available - device has SMART capability.
SMART support is:      Enabled
Temperature Warning:   Enabled
Read Cache is:         Enabled
Writeback Cache is:    Disabled

=== START OF READ SMART DATA SECTION ===
SMART Health Status: OK

Percentage used endurance indicator: 0%
Current temperature = 34
Lifetime maximum temperature = 38
Lifetime minimum temperature = 7
Maximum temperature since power on = 34
Minimum temperature since power on = 16
Manufactured in week 46 of year 2021
Accumulated start-stop cycles: 73
Specified load-unload count over device lifetime: 0
Accumulated load-unload cycles: 0
Grown defect list length=1 bytes [unknown number of elements]

Error counter log:
      Errors Corrected by      Total   Correction   Gigabytes   Total
      ECC      by      rereads/   errors   algorithm   processed   uncorrected
      fast | delayed  rewrites  corrected  invocations [10^9 bytes]  errors
read:          0         0          0         0         0         0.000         0
write:         0         0          0         0         0         0.000         0

Non-medium error count:      1905

Pending defect count:0 Pending Defects
SMART Self-test log
Num  Test              Status              segment  LifeTime  LBA_first_err [SK ASC ASQ]
     Description
# 1  Background short  Completed              -        145              - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]

Background scan results log
Status: scan is active
Accumulated power on time, hours:minutes 164:47 [9887 minutes]
Number of background scans performed: 0, scan progress: 15.41%
Number of background medium scans performed: 0
Device does not support General statistics and performance logging

Protocol Specific port log page for SAS SSP
relative target port id = 1
```

```
generation code = 2
number of phys = 1
phy identifier = 0
  attached device type: SAS or SATA device
  attached reason: loss of dword synchronization
  reason: loss of dword synchronization
  negotiated logical link rate: phy enabled; 12 Gbps
  attached initiator port: ssp=1 stp=1 smp=1
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b65132
  attached SAS address = 0x500304802485ab01
  attached phy identifier = 0
  Invalid DWORD count = 0
  Running disparity error count = 0
  Loss of DWORD synchronization count = 0
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0
relative target port id = 2
generation code = 2
number of phys = 1
phy identifier = 1
  attached device type: no device attached
  attached reason: unknown
  reason: unknown
  negotiated logical link rate: phy enabled; unknown
  attached initiator port: ssp=0 stp=0 smp=0
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b65133
  attached SAS address = 0x0
  attached phy identifier = 0
  Invalid DWORD count = 0
  Running disparity error count = 0
  Loss of DWORD synchronization count = 0
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0

===== Selbsttestprotokoll =====
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=== START OF READ SMART DATA SECTION ===
SMART Self-test log
Num Test Status segment LifeTime LBA_first_err [SK ASC ASQ]
Description number (hours)
# 1 Background short Completed - 145 - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]
```

S5KUNA0RB02923

```
#####
# Datentraeger /dev/sda Seriennummer S5KUNA0RB02923
# Erhoben am 2026-08-27 23:27:18 CEST auf Supermicro X13SAE-F
#####

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
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=== START OF INFORMATION SECTION ===
Vendor:                HPE
Product:               V0003200JWZJR
Revision:              HPD5
Compliance:           SPC-5
User Capacity:         3,200,631,791,616 bytes [3.20 TB]
Logical block size:    512 bytes
Physical block size:   4096 bytes
LU is resource provisioned, LBPRZ=1
Rotation Rate:        Solid State Device
Form Factor:          2.5 inches
Logical Unit id:       0x50000f0b01b653a0
Serial number:        S5KUNA0RB02923
Device type:          disk
Transport protocol:    SAS (SPL-4)
Local Time is:        Thu Aug 27 23:27:18 2026 CEST
SMART support is:      Available - device has SMART capability.
SMART support is:      Enabled
Temperature Warning:   Enabled
Read Cache is:         Enabled
Writeback Cache is:    Disabled

=== START OF READ SMART DATA SECTION ===
SMART Health Status: OK

Percentage used endurance indicator: 0%
Current temperature = 34
Lifetime maximum temperature = 34
Lifetime minimum temperature = 9
Maximum temperature since power on = 34
Minimum temperature since power on = 26
Manufactured in week 46 of year 2021
Accumulated start-stop cycles: 48
Specified load-unload count over device lifetime: 0
Accumulated load-unload cycles: 0
Grown defect list length=1 bytes [unknown number of elements]

Error counter log:
      Errors Corrected by      Total   Correction   Gigabytes   Total
      ECC      by      reads/   errors   algorithm   processed   uncorrected
      fast | delayed  rewrites corrected invocations [10^9 bytes] errors
read:         0         0         0         0         0         0.000         0
write:        0         0         0         0         0         0.000         0

Non-medium error count:         4288

Pending defect count:0 Pending Defects
SMART Self-test log
Num  Test              Status      segment  LifeTime  LBA_first_err [SK ASC ASQ]
     Description
# 1  Background short  Completed          -      196              - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]

Background scan results log
Status: scan is active
Accumulated power on time, hours:minutes 196:59 [11819 minutes]
Number of background scans performed: 0, scan progress: 24.90%
Number of background medium scans performed: 0
Device does not support General statistics and performance logging

Protocol Specific port log page for SAS SSP
relative target port id = 1
```

```
generation code = 10
number of phys = 1
phy identifier = 0
  attached device type: SAS or SATA device
  attached reason: power on
  reason: loss of dword synchronization
  negotiated logical link rate: phy enabled; 12 Gbps
  attached initiator port: ssp=1 stp=1 smp=1
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b653a2
  attached SAS address = 0x500304802485ab01
  attached phy identifier = 0
  Invalid DWORD count = 43
  Running disparity error count = 43
  Loss of DWORD synchronization count = 8
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0
relative target port id = 2
generation code = 10
number of phys = 1
phy identifier = 1
  attached device type: no device attached
  attached reason: unknown
  reason: unknown
  negotiated logical link rate: phy enabled; unknown
  attached initiator port: ssp=0 stp=0 smp=0
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b653a3
  attached SAS address = 0x0
  attached phy identifier = 0
  Invalid DWORD count = 0
  Running disparity error count = 0
  Loss of DWORD synchronization count = 0
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0

===== Selbsttestprotokoll =====
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=== START OF READ SMART DATA SECTION ===
SMART Self-test log
Num Test Status segment LifeTime LBA_first_err [SK ASC ASQ]
Description number (hours)
# 1 Background short Completed - 196 - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]
```

S5KUNA0RB02929

```
#####
# Datentraeger /dev/sdd Seriennummer S5KUNA0RB02929
# Erhoben am 2026-08-28 19:20:54 CEST auf Supermicro X13SAE-F
#####

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
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=== START OF INFORMATION SECTION ===
Vendor:                HPE
Product:               V0003200JWZJR
Revision:              HPD5
Compliance:            SPC-5
User Capacity:         3,200,631,791,616 bytes [3.20 TB]
Logical block size:    512 bytes
Physical block size:   4096 bytes
LU is resource provisioned, LBPRZ=1
Rotation Rate:         Solid State Device
Form Factor:           2.5 inches
Logical Unit id:       0x50000f0b01b65400
Serial number:         S5KUNA0RB02929
Device type:           disk
Transport protocol:    SAS (SPL-4)
Local Time is:         Fri Aug 28 19:20:54 2026 CEST
SMART support is:      Available - device has SMART capability.
SMART support is:      Enabled
Temperature Warning:   Enabled
Read Cache is:         Enabled
Writeback Cache is:    Disabled

=== START OF READ SMART DATA SECTION ===
SMART Health Status: OK

Percentage used endurance indicator: 0%
Current temperature = 35
Lifetime maximum temperature = 36
Lifetime minimum temperature = 9
Maximum temperature since power on = 36
Minimum temperature since power on = 26
Manufactured in week 46 of year 2021
Accumulated start-stop cycles: 48
Specified load-unload count over device lifetime: 0
Accumulated load-unload cycles: 0
Grown defect list length=1 bytes [unknown number of elements]

Error counter log:
      Errors Corrected by           Total   Correction   Gigabytes   Total
      ECC      rewrites/    errors    algorithm   processed   uncorrected
      fast | delayed   rewrites   corrected invocations [10^9 bytes]   errors
read:           0         0           0           0           0.000           0
write:          0         0           0           0           0.000           0

Non-medium error count:           2363

Pending defect count:0 Pending Defects
SMART Self-test log
Num  Test              Status              segment  LifeTime  LBA_first_err [SK ASC ASQ]
     Description
# 1  Background short  Completed              -        196              - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]

Background scan results log
Status: scan is active
Accumulated power on time, hours:minutes 216:56 [13016 minutes]
Number of background scans performed: 0, scan progress: 27.82%
Number of background medium scans performed: 0
Device does not support General statistics and performance logging

Protocol Specific port log page for SAS SSP
relative target port id = 1
```

```
generation code = 19
number of phys = 1
phy identifier = 0
  attached device type: SAS or SATA device
  attached reason: loss of dword synchronization
  reason: loss of dword synchronization
  negotiated logical link rate: phy enabled; 12 Gbps
  attached initiator port: ssp=1 stp=1 smp=1
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b65402
  attached SAS address = 0x500304802485ab07
  attached phy identifier = 6
  Invalid DWORD count = 81
  Running disparity error count = 81
  Loss of DWORD synchronization count = 17
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0
relative target port id = 2
generation code = 19
number of phys = 1
phy identifier = 1
  attached device type: no device attached
  attached reason: unknown
  reason: unknown
  negotiated logical link rate: phy enabled; unknown
  attached initiator port: ssp=0 stp=0 smp=0
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b65403
  attached SAS address = 0x0
  attached phy identifier = 0
  Invalid DWORD count = 0
  Running disparity error count = 0
  Loss of DWORD synchronization count = 0
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0

===== Selbsttestprotokoll =====
smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
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=== START OF READ SMART DATA SECTION ===
SMART Self-test log
Num  Test          Status          segment  LifeTime  LBA_first_err [SK ASC ASQ]
   1  Description
# 1  Background short Completed        -      196             - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]
```

S5KUNA0RB02935

```
#####
# Datentraeger /dev/sdm   Seriennummer S5KUNA0RB02935
# Erhoben am 2026-08-28 19:21:05 CEST auf Supermicro X13SAE-F
#####

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
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=== START OF INFORMATION SECTION ===
Vendor:                HPE
Product:               V0003200JWZJR
Revision:              HPD5
Compliance:            SPC-5
User Capacity:         3,200,631,791,616 bytes [3.20 TB]
Logical block size:    512 bytes
Physical block size:   4096 bytes
LU is resource provisioned, LBPRZ=1
Rotation Rate:         Solid State Device
Form Factor:           2.5 inches
Logical Unit id:       0x50000f0b01b65460
Serial number:         S5KUNA0RB02935
Device type:           disk
Transport protocol:    SAS (SPL-4)
Local Time is:         Fri Aug 28 19:21:05 2026 CEST
SMART support is:      Available - device has SMART capability.
SMART support is:      Enabled
Temperature Warning:   Enabled
Read Cache is:         Enabled
Writeback Cache is:    Disabled

=== START OF READ SMART DATA SECTION ===
SMART Health Status: OK

Percentage used endurance indicator: 0%
Current temperature = 35
Lifetime maximum temperature = 35
Lifetime minimum temperature = 9
Maximum temperature since power on = 35
Minimum temperature since power on = 26
Manufactured in week 46 of year 2021
Accumulated start-stop cycles: 48
Specified load-unload count over device lifetime: 0
Accumulated load-unload cycles: 0
Grown defect list length=1 bytes [unknown number of elements]

Error counter log:
      Errors Corrected by      Total   Correction   Gigabytes   Total
      ECC      by      rereads/   errors   algorithm   processed   uncorrected
      fast | delayed  rewrites  corrected  invocations [10^9 bytes]  errors
read:          0         0          0         0         0         0.000         0
write:         0         0          0         0         0         0.000         0

Non-medium error count:      2362

Pending defect count:0 Pending Defects
SMART Self-test log
Num  Test              Status              segment  LifeTime  LBA_first_err [SK ASC ASQ]
     Description
# 1 Background short Completed              -        196              - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]

Background scan results log
Status: scan is active
Accumulated power on time, hours:minutes 216:56 [13016 minutes]
Number of background scans performed: 0, scan progress: 27.83%
Number of background medium scans performed: 0
Device does not support General statistics and performance logging

Protocol Specific port log page for SAS SSP
relative target port id = 1
```

```
generation code = 19
number of phys = 1
phy identifier = 0
  attached device type: SAS or SATA device
  attached reason: loss of dword synchronization
  reason: loss of dword synchronization
  negotiated logical link rate: phy enabled; 12 Gbps
  attached initiator port: ssp=1 stp=1 smp=1
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b65462
  attached SAS address = 0x500304802485ab04
  attached phy identifier = 3
  Invalid DWORD count = 85
  Running disparity error count = 83
  Loss of DWORD synchronization count = 17
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0
relative target port id = 2
generation code = 19
number of phys = 1
phy identifier = 1
  attached device type: no device attached
  attached reason: unknown
  reason: unknown
  negotiated logical link rate: phy enabled; unknown
  attached initiator port: ssp=0 stp=0 smp=0
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b65463
  attached SAS address = 0x0
  attached phy identifier = 0
  Invalid DWORD count = 0
  Running disparity error count = 0
  Loss of DWORD synchronization count = 0
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0

===== Selbsttestprotokoll =====
smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
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=== START OF READ SMART DATA SECTION ===
SMART Self-test log
Num Test Status segment LifeTime LBA_first_err [SK ASC ASQ]
Description number (hours)
# 1 Background short Completed - 196 - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]
```

S5KUNA0RB02886

```
#####
# Datentraeger /dev/sdf Seriennummer S5KUNA0RB02886
# Erhoben am 2026-08-27 23:27:24 CEST auf Supermicro X13SAE-F
#####

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF INFORMATION SECTION ===
Vendor:                HPE
Product:               V0003200JWZJR
Revision:              HPD5
Compliance:            SPC-5
User Capacity:         3,200,631,791,616 bytes [3.20 TB]
Logical block size:    512 bytes
Physical block size:   4096 bytes
LU is resource provisioned, LBPRZ=1
Rotation Rate:         Solid State Device
Form Factor:           2.5 inches
Logical Unit id:        0x50000f0b01b65150
Serial number:          S5KUNA0RB02886
Device type:           disk
Transport protocol:    SAS (SPL-4)
Local Time is:         Thu Aug 27 23:27:24 2026 CEST
SMART support is:      Available - device has SMART capability.
SMART support is:      Enabled
Temperature Warning:   Enabled
Read Cache is:         Enabled
Writeback Cache is:    Disabled

=== START OF READ SMART DATA SECTION ===
SMART Health Status: OK

Percentage used endurance indicator: 0%
Current temperature = 34
Lifetime maximum temperature = 34
Lifetime minimum temperature = 12
Maximum temperature since power on = 34
Minimum temperature since power on = 26
Manufactured in week 46 of year 2021
Accumulated start-stop cycles: 47
Specified load-unload count over device lifetime: 0
Accumulated load-unload cycles: 0
Grown defect list length=1 bytes [unknown number of elements]

Error counter log:
           Errors Corrected by           Total   Correction   Gigabytes   Total
           ECC      by rereads/         errors   algorithm   processed   uncorrected
           fast | delayed  rewrites   corrected invocations [10^9 bytes]   errors
read:           0         0             0           0           0.000           0
write:          0         0             0           0           0.000           0

Non-medium error count:           2332

Pending defect count:0 Pending Defects
SMART Self-test log
Num  Test           Status              segment  LifeTime  LBA_first_err [SK ASC ASQ]
     Description
# 1 Background short Completed              -        198             - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]

Background scan results log
Status: scan is active
Accumulated power on time, hours:minutes 198:05 [11885 minutes]
Number of background scans performed: 0, scan progress: 24.90%
Number of background medium scans performed: 0
Device does not support General statistics and performance logging

Protocol Specific port log page for SAS SSP
relative target port id = 1
```

```
generation code = 19
number of phys = 1
phy identifier = 0
  attached device type: SAS or SATA device
  attached reason: unknown
  reason: loss of dword synchronization
  negotiated logical link rate: phy enabled; 12 Gbps
  attached initiator port: ssp=1 stp=1 smp=1
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b65152
  attached SAS address = 0x500304802485ab03
  attached phy identifier = 2
  Invalid DWORD count = 79
  Running disparity error count = 78
  Loss of DWORD synchronization count = 17
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0
relative target port id = 2
generation code = 19
number of phys = 1
phy identifier = 1
  attached device type: no device attached
  attached reason: unknown
  reason: unknown
  negotiated logical link rate: phy enabled; unknown
  attached initiator port: ssp=0 stp=0 smp=0
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b65153
  attached SAS address = 0x0
  attached phy identifier = 0
  Invalid DWORD count = 0
  Running disparity error count = 0
  Loss of DWORD synchronization count = 0
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0

===== Selbsttestprotokoll =====
smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
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=== START OF READ SMART DATA SECTION ===
SMART Self-test log
Num  Test          Status      segment  LifeTime  LBA_first_err [SK ASC ASQ]
   1  Description
# 1  Background short Completed    -        198             - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]
```

S5KUNA0RB02911

```
#####
# Datentraeger /dev/sdk Seriennummer S5KUNA0RB02911
# Erhoben am 2026-08-28 19:21:02 CEST auf Supermicro X13SAE-F
#####

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF INFORMATION SECTION ===
Vendor:                HPE
Product:               V0003200JWZJR
Revision:              HPD5
Compliance:            SPC-5
User Capacity:         3,200,631,791,616 bytes [3.20 TB]
Logical block size:    512 bytes
Physical block size:   4096 bytes
LU is resource provisioned, LBPRZ=1
Rotation Rate:         Solid State Device
Form Factor:           2.5 inches
Logical Unit id:       0x50000f0b01b652e0
Serial number:         S5KUNA0RB02911
Device type:           disk
Transport protocol:    SAS (SPL-4)
Local Time is:         Fri Aug 28 19:21:02 2026 CEST
SMART support is:      Available - device has SMART capability.
SMART support is:      Enabled
Temperature Warning:   Enabled
Read Cache is:         Enabled
Writeback Cache is:    Disabled

=== START OF READ SMART DATA SECTION ===
SMART Health Status: OK

Percentage used endurance indicator: 0%
Current temperature = 34
Lifetime maximum temperature = 40
Lifetime minimum temperature = 9
Maximum temperature since power on = 35
Minimum temperature since power on = 26
Manufactured in week 46 of year 2021
Accumulated start-stop cycles: 71
Specified load-unload count over device lifetime: 0
Accumulated load-unload cycles: 0
Grown defect list length=1 bytes [unknown number of elements]

Error counter log:
      Errors Corrected by      Total   Correction   Gigabytes   Total
      ECC      by      rereads/   errors   algorithm   processed   uncorrected
      fast | delayed  rewrites  corrected  invocations [10^9 bytes]  errors
read:          0         0          0         0         0         0.000         0
write:         0         0          0         0         0         0.000         0

Non-medium error count:         2712

Pending defect count:0 Pending Defects
SMART Self-test log
Num  Test              Status              segment  LifeTime  LBA_first_err [SK ASC ASQ]
     Description
# 1  Background short  Completed           -         303             - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]

Background scan results log
Status: scan is active
Accumulated power on time, hours:minutes 323:31 [19411 minutes]
Number of background scans performed: 0, scan progress: 27.83%
Number of background medium scans performed: 0
Device does not support General statistics and performance logging

Protocol Specific port log page for SAS SSP
relative target port id = 1
```

```
generation code = 19
number of phys = 1
phy identifier = 0
  attached device type: SAS or SATA device
  attached reason: loss of dword synchronization
  reason: loss of dword synchronization
  negotiated logical link rate: phy enabled; 12 Gbps
  attached initiator port: ssp=1 stp=1 smp=1
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b652e2
  attached SAS address = 0x500304802485ab06
  attached phy identifier = 5
  Invalid DWORD count = 80
  Running disparity error count = 80
  Loss of DWORD synchronization count = 17
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0
relative target port id = 2
generation code = 19
number of phys = 1
phy identifier = 1
  attached device type: no device attached
  attached reason: unknown
  reason: unknown
  negotiated logical link rate: phy enabled; unknown
  attached initiator port: ssp=0 stp=0 smp=0
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b652e3
  attached SAS address = 0x0
  attached phy identifier = 0
  Invalid DWORD count = 0
  Running disparity error count = 0
  Loss of DWORD synchronization count = 0
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0

===== Selbsttestprotokoll =====
smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
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=== START OF READ SMART DATA SECTION ===
SMART Self-test log
Num  Test          Status      segment  LifeTime  LBA_first_err [SK ASC ASQ]
   1  Description
# 1  Background short Completed    -        303             - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]
```

S5KUNA0RB02930

```
#####
# Datentraeger /dev/sdc Seriennummer S5KUNA0RB02930
# Erhoben am 2026-08-28 19:20:53 CEST auf Supermicro X13SAE-F
#####

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
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=== START OF INFORMATION SECTION ===
Vendor:                HPE
Product:               V0003200JWZJR
Revision:              HPD5
Compliance:            SPC-5
User Capacity:         3,200,631,791,616 bytes [3.20 TB]
Logical block size:    512 bytes
Physical block size:   4096 bytes
LU is resource provisioned, LBPRZ=1
Rotation Rate:         Solid State Device
Form Factor:           2.5 inches
Logical Unit id:       0x50000f0b01b65410
Serial number:         S5KUNA0RB02930
Device type:           disk
Transport protocol:    SAS (SPL-4)
Local Time is:         Fri Aug 28 19:20:53 2026 CEST
SMART support is:      Available - device has SMART capability.
SMART support is:      Enabled
Temperature Warning:   Enabled
Read Cache is:         Enabled
Writeback Cache is:    Disabled

=== START OF READ SMART DATA SECTION ===
SMART Health Status: OK

Percentage used endurance indicator: 0%
Current temperature = 35
Lifetime maximum temperature = 40
Lifetime minimum temperature = 8
Maximum temperature since power on = 35
Minimum temperature since power on = 27
Manufactured in week 46 of year 2021
Accumulated start-stop cycles: 76
Specified load-unload count over device lifetime: 0
Accumulated load-unload cycles: 0
Grown defect list length=1 bytes [unknown number of elements]

Error counter log:
      Errors Corrected by      Total   Correction   Gigabytes   Total
      ECC      by      rereads/   errors   algorithm   processed   uncorrected
      fast | delayed  rewrites  corrected  invocations [10^9 bytes]  errors
read:         0         0         0         0         0         0.000         0
write:        0         0         0         0         0         0.000         0

Non-medium error count:      3929

Pending defect count:0 Pending Defects
SMART Self-test log
Num  Test              Status              segment  LifeTime  LBA_first_err [SK ASC ASQ]
     Description
# 1  Background short  Completed              -        314              - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]

Background scan results log
Status: scan is active
Accumulated power on time, hours:minutes 334:50 [20090 minutes]
Number of background scans performed: 0, scan progress: 27.81%
Number of background medium scans performed: 0
Device does not support General statistics and performance logging

Protocol Specific port log page for SAS SSP
relative target port id = 1
```

```
generation code = 19
number of phys = 1
phy identifier = 0
  attached device type: SAS or SATA device
  attached reason: unknown
  reason: loss of dword synchronization
  negotiated logical link rate: phy enabled; 12 Gbps
  attached initiator port: ssp=1 stp=1 smp=1
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b65412
  attached SAS address = 0x500304802485ab02
  attached phy identifier = 1
  Invalid DWORD count = 79
  Running disparity error count = 78
  Loss of DWORD synchronization count = 17
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0
relative target port id = 2
generation code = 19
number of phys = 1
phy identifier = 1
  attached device type: no device attached
  attached reason: unknown
  reason: unknown
  negotiated logical link rate: phy enabled; unknown
  attached initiator port: ssp=0 stp=0 smp=0
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b65413
  attached SAS address = 0x0
  attached phy identifier = 0
  Invalid DWORD count = 0
  Running disparity error count = 0
  Loss of DWORD synchronization count = 0
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0

===== Selbsttestprotokoll =====
smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
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=== START OF READ SMART DATA SECTION ===
SMART Self-test log
Num Test          Status      segment  LifeTime  LBA_first_err [SK ASC ASQ]
   #  Description                number    (hours)
#  1  Background short Completed        -      314             - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]
```

S5KUNA0RB03018

```
#####
# Datentraeger /dev/sdi Seriennummer S5KUNA0RB03018
# Erhoben am 2026-08-28 19:21:00 CEST auf Supermicro X13SAE-F
#####

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF INFORMATION SECTION ===
Vendor:                HPE
Product:               V0003200JWZJR
Revision:              HPD5
Compliance:            SPC-5
User Capacity:         3,200,631,791,616 bytes [3.20 TB]
Logical block size:    512 bytes
Physical block size:   4096 bytes
LU is resource provisioned, LBPRZ=1
Rotation Rate:         Solid State Device
Form Factor:           2.5 inches
Logical Unit id:       0x50000f0b01b65990
Serial number:         S5KUNA0RB03018
Device type:           disk
Transport protocol:    SAS (SPL-4)
Local Time is:         Fri Aug 28 19:21:00 2026 CEST
SMART support is:      Available - device has SMART capability.
SMART support is:      Enabled
Temperature Warning:   Enabled
Read Cache is:         Enabled
Writeback Cache is:    Disabled

=== START OF READ SMART DATA SECTION ===
SMART Health Status: OK

Percentage used endurance indicator: 0%
Current temperature = 34
Lifetime maximum temperature = 39
Lifetime minimum temperature = 9
Maximum temperature since power on = 35
Minimum temperature since power on = 26
Manufactured in week 46 of year 2021
Accumulated start-stop cycles: 72
Specified load-unload count over device lifetime: 0
Accumulated load-unload cycles: 0
Grown defect list length=1 bytes [unknown number of elements]

Error counter log:
      Errors Corrected by      Total   Correction   Gigabytes   Total
      ECC      by      rereads/   errors   algorithm   processed   uncorrected
      fast | delayed  rewrites  corrected  invocations [10^9 bytes]  errors
read:          0         0          0         0         0         0.000         0
write:         0         0          0         0         0         0.000         0

Non-medium error count:         2748

Pending defect count:0 Pending Defects
SMART Self-test log
Num  Test              Status              segment  LifeTime  LBA_first_err [SK ASC ASQ]
     Description
# 1 Background short  Completed          -         364          - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]

Background scan results log
Status: scan is active
Accumulated power on time, hours:minutes 384:37 [23077 minutes]
Number of background scans performed: 0, scan progress: 27.81%
Number of background medium scans performed: 0
Device does not support General statistics and performance logging

Protocol Specific port log page for SAS SSP
relative target port id = 1
```

```
generation code = 19
number of phys = 1
phy identifier = 0
  attached device type: SAS or SATA device
  attached reason: loss of dword synchronization
  reason: loss of dword synchronization
  negotiated logical link rate: phy enabled; 12 Gbps
  attached initiator port: ssp=1 stp=1 smp=1
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b65992
  attached SAS address = 0x500304802485ab05
  attached phy identifier = 4
  Invalid DWORD count = 80
  Running disparity error count = 80
  Loss of DWORD synchronization count = 17
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0
relative target port id = 2
generation code = 19
number of phys = 1
phy identifier = 1
  attached device type: no device attached
  attached reason: unknown
  reason: unknown
  negotiated logical link rate: phy enabled; unknown
  attached initiator port: ssp=0 stp=0 smp=0
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b65993
  attached SAS address = 0x0
  attached phy identifier = 0
  Invalid DWORD count = 0
  Running disparity error count = 0
  Loss of DWORD synchronization count = 0
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0

===== Selbsttestprotokoll =====
smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
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=== START OF READ SMART DATA SECTION ===
SMART Self-test log
Num  Test          Status      segment  LifeTime  LBA_first_err [SK ASC ASQ]
   1  Description
# 1  Background short Completed    -        364             - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]
```

S5KUNA0RB03024

```
#####
# Datentraeger /dev/sdh   Seriennummer S5KUNA0RB03024
# Erhoben am 2026-08-28 19:20:59 CEST auf Supermicro X13SAE-F
#####

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF INFORMATION SECTION ===
Vendor:                HPE
Product:               V0003200JWZJR
Revision:              HPD5
Compliance:            SPC-5
User Capacity:         3,200,631,791,616 bytes [3.20 TB]
Logical block size:    512 bytes
Physical block size:   4096 bytes
LU is resource provisioned, LBPRZ=1
Rotation Rate:         Solid State Device
Form Factor:           2.5 inches
Logical Unit id:        0x50000f0b01b659f0
Serial number:         S5KUNA0RB03024
Device type:           disk
Transport protocol:    SAS (SPL-4)
Local Time is:         Fri Aug 28 19:20:59 2026 CEST
SMART support is:      Available - device has SMART capability.
SMART support is:      Enabled
Temperature Warning:   Enabled
Read Cache is:         Enabled
Writeback Cache is:    Disabled

=== START OF READ SMART DATA SECTION ===
SMART Health Status: OK

Percentage used endurance indicator: 0%
Current temperature = 35
Lifetime maximum temperature = 38
Lifetime minimum temperature = 7
Maximum temperature since power on = 36
Minimum temperature since power on = 27
Manufactured in week 46 of year 2021
Accumulated start-stop cycles: 72
Specified load-unload count over device lifetime: 0
Accumulated load-unload cycles: 0
Grown defect list length=1 bytes [unknown number of elements]

Error counter log:
      Errors Corrected by      Total   Correction   Gigabytes   Total
      ECC      by      rereads/   errors   algorithm   processed   uncorrected
      fast | delayed  rewrites  corrected  invocations [10^9 bytes]  errors
read:          0         0          0         0         0         0.000         0
write:         0         0          0         0         0         0.000         0

Non-medium error count:      3933

Pending defect count:0 Pending Defects
SMART Self-test log
Num  Test              Status              segment  LifeTime  LBA_first_err [SK ASC ASQ]
     Description
# 1  Background short  Completed              -        375              - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]

Background scan results log
Status: scan is active
Accumulated power on time, hours:minutes 395:01 [23701 minutes]
Number of background scans performed: 0, scan progress: 27.81%
Number of background medium scans performed: 0
Device does not support General statistics and performance logging

Protocol Specific port log page for SAS SSP
relative target port id = 1
```

```
generation code = 19
number of phys = 1
phy identifier = 0
  attached device type: SAS or SATA device
  attached reason: loss of dword synchronization
  reason: loss of dword synchronization
  negotiated logical link rate: phy enabled; 12 Gbps
  attached initiator port: ssp=1 stp=1 smp=1
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b659f2
  attached SAS address = 0x500304802485ab08
  attached phy identifier = 7
  Invalid DWORD count = 80
  Running disparity error count = 80
  Loss of DWORD synchronization count = 17
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0
relative target port id = 2
generation code = 19
number of phys = 1
phy identifier = 1
  attached device type: no device attached
  attached reason: unknown
  reason: unknown
  negotiated logical link rate: phy enabled; unknown
  attached initiator port: ssp=0 stp=0 smp=0
  attached target port: ssp=0 stp=0 smp=0
  SAS address = 0x50000f0b01b659f3
  attached SAS address = 0x0
  attached phy identifier = 0
  Invalid DWORD count = 0
  Running disparity error count = 0
  Loss of DWORD synchronization count = 0
  Phy reset problem count = 0
  Phy event descriptors:
    Received SSP frame error count: 0

===== Selbsttestprotokoll =====
smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF READ SMART DATA SECTION ===
SMART Self-test log
Num  Test          Status      segment  LifeTime  LBA_first_err [SK ASC ASQ]
   1  Description
# 1  Background short Completed    -      375             - [- - -]

Long (extended) Self-test duration: 1634 seconds [27.2 minutes]
```

ZA2DN4VV

```
#####
# Datentraeger /dev/sdn Seriennummer ZA2DN4VV
# Erhoben am 2026-08-28 19:21:06 CEST auf Supermicro X13SAE-F
#####

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF INFORMATION SECTION ===
Model Family:      Seagate Enterprise Capacity 3.5 HDD
Device Model:      ST10000NM0086-2AA101
Serial Number:     ZA2DN4VV
LU WWN Device Id:  5 000c50 0c6f24439
Firmware Version:  SN05
User Capacity:     10,000,831,348,736 bytes [10.0 TB]
Sector Sizes:      512 bytes logical, 4096 bytes physical
Rotation Rate:     7200 rpm
Form Factor:       3.5 inches
Device is:         In smartctl database 7.3/5528
ATA Version is:    ACS-3 T13/2161-D revision 5
SATA Version is:   SATA 3.1, 6.0 Gb/s (current: 6.0 Gb/s)
Local Time is:     Fri Aug 28 19:21:06 2026 CEST
SMART support is:  Available - device has SMART capability.
SMART support is:  Enabled
AAM feature is:    Unavailable
APM feature is:    Unavailable
Rd look-ahead is:  Enabled
Write cache is:    Enabled
DSN feature is:    Unavailable
ATA Security is:   Disabled, NOT FROZEN [SEC1]
Wt Cache Reorder:  Enabled

=== START OF READ SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

General SMART Values:
Offline data collection status:  (0x82) Offline data collection activity
                                   was completed without error.
                                   Auto Offline Data Collection: Enabled.
Self-test execution status:      (   0) The previous self-test routine completed
                                   without error or no self-test has ever
                                   been run.

Total time to complete Offline
data collection:                  ( 575) seconds.
Offline data collection
capabilities:                      (0x7b) SMART execute Offline immediate.
                                   Auto Offline data collection on/off support.
                                   Suspend Offline collection upon new
                                   command.
                                   Offline surface scan supported.
                                   Self-test supported.
                                   Conveyance Self-test supported.
                                   Selective Self-test supported.
SMART capabilities:              (0x0003) Saves SMART data before entering
                                   power-saving mode.
                                   Supports SMART auto save timer.
Error logging capability:        (0x01) Error logging supported.
                                   General Purpose Logging supported.

Short self-test routine
recommended polling time:         (   1) minutes.
Extended self-test routine
recommended polling time:         (  96) minutes.
Conveyance self-test routine
recommended polling time:         (   2) minutes.
SCT capabilities:                (0x70bd) SCT Status supported.
                                   SCT Error Recovery Control supported.
                                   SCT Feature Control supported.
                                   SCT Data Table supported.

SMART Attributes Data Structure revision number: 10
Vendor Specific SMART Attributes with Thresholds:
```

ID#	ATTRIBUTE_NAME	FLAGS	VALUE	WORST	THRESH	FAIL	RAW_VALUE
1	Raw_Read_Error_Rate	POSR--	082	064	044	-	151094834
3	Spin_Up_Time	P0----	093	091	000	-	0
4	Start_Stop_Count	-0--CK	100	100	020	-	866
5	Reallocated_Sector_Ct	P0--CK	100	100	010	-	0
7	Seek_Error_Rate	POSR--	087	060	045	-	536977017
9	Power_On_Hours	-0--CK	096	096	000	-	4363
10	Spin_Retry_Count	P0--C-	100	100	097	-	0
12	Power_Cycle_Count	-0--CK	100	100	020	-	866
184	End-to-End_Error	-0--CK	100	100	099	-	0
187	Reported_Uncorrect	-0--CK	100	100	000	-	0
188	Command_Timeout	-0--CK	100	099	000	-	0 0 2
189	High_Fly_Writes	-0-RCK	041	041	000	-	59
190	Airflow_Temperature_Cel	-0---K	062	053	040	-	38 (Min/Max 30/43)
191	G-Sense_Error_Rate	-0--CK	098	098	000	-	4515
192	Power-Off_Retract_Count	-0--CK	100	100	000	-	861
193	Load_Cycle_Count	-0--CK	097	097	000	-	6730
194	Temperature_Celsius	-0---K	038	047	000	-	38 (0 19 0 0 0)
195	Hardware_ECC_Recovered	-0-RC-	082	064	000	-	151094834
197	Current_Pending_Sector	-0--C-	100	100	000	-	0
198	Offline_Uncorrectable	----C-	100	100	000	-	0
199	UDMA_CRC_Error_Count	-OSRCK	200	200	000	-	0
200	Pressure_Limit	P0---K	100	100	001	-	0
240	Head_Flying_Hours	-----	100	253	000	-	4094h+34m+43.063s
241	Total_LBAs_Written	-----	100	253	000	-	65239630232
242	Total_LBAs_Read	-----	100	253	000	-	118093618585
		_	K auto-keep				
		_	C event count				
		_	R error rate				
		_ _	S speed/performance				
		_ _ _	O updated online				
		_ _ _	P prefailure warning				

General Purpose Log Directory Version 1

SMART Log Directory Version 1 [multi-sector log support]

Address	Access	R/W	Size	Description
0x00	GPL,SL	R/O	1	Log Directory
0x01	SL	R/O	1	Summary SMART error log
0x02	SL	R/O	5	Comprehensive SMART error log
0x03	GPL	R/O	5	Ext. Comprehensive SMART error log
0x04	GPL	R/O	256	Device Statistics log
0x04	SL	R/O	8	Device Statistics log
0x06	SL	R/O	1	SMART self-test log
0x07	GPL	R/O	1	Extended self-test log
0x08	GPL	R/O	2	Power Conditions log
0x09	SL	R/W	1	Selective self-test log
0x0c	GPL	R/O	2048	Pending Defects log
0x10	GPL	R/O	1	NCQ Command Error log
0x11	GPL	R/O	1	SATA Phy Event Counters log
0x13	GPL	R/O	1	SATA NCQ Send and Receive log
0x15	GPL	R/W	1	Rebuild Assist log
0x21	GPL	R/O	1	Write stream error log
0x22	GPL	R/O	1	Read stream error log
0x24	GPL	R/O	768	Current Device Internal Status Data log
0x30	GPL,SL	R/O	9	IDENTIFY DEVICE data log
0x80-0x9f	GPL,SL	R/W	16	Host vendor specific log
0xa1	GPL,SL	VS	24	Device vendor specific log
0xa2	GPL	VS	16320	Device vendor specific log
0xa6	GPL	VS	192	Device vendor specific log
0xa8-0xa9	GPL,SL	VS	136	Device vendor specific log
0xab	GPL	VS	1	Device vendor specific log
0xad	GPL	VS	16	Device vendor specific log
0xb0	GPL	VS	17208	Device vendor specific log
0xbe-0xbf	GPL	VS	65535	Device vendor specific log
0xc1	GPL,SL	VS	16	Device vendor specific log
0xc3	GPL,SL	VS	56	Device vendor specific log
0xc9	GPL,SL	VS	8	Device vendor specific log
0xca	GPL,SL	VS	16	Device vendor specific log
0xd1	GPL	VS	304	Device vendor specific log
0xd2	GPL	VS	10000	Device vendor specific log
0xd4	GPL	VS	2048	Device vendor specific log
0xe0	GPL,SL	R/W	1	SCT Command/Status
0xe1	GPL,SL	R/W	1	SCT Data Transfer

SMART Extended Comprehensive Error Log Version: 1 (5 sectors)

No Errors Logged

SMART Extended Self-test Log Version: 1 (1 sectors)

Num	Test_Description	Status	Remaining	LifeTime(hours)	LBA_of_first_error
# 1	Extended offline	Completed without error	00%	4358	-
# 2	Short offline	Completed without error	00%	4343	-
# 3	Short offline	Completed without error	00%	4060	-
# 4	Extended offline	Interrupted (host reset)	60%	4056	-
# 5	Short offline	Interrupted (host reset)	00%	4035	-
# 6	Short offline	Completed without error	00%	4017	-
# 7	Extended offline	Completed without error	00%	3998	-
# 8	Short offline	Completed without error	00%	3969	-
# 9	Short offline	Completed without error	00%	3959	-
#10	Short offline	Completed without error	00%	3938	-
#11	Short offline	Completed without error	00%	3914	-
#12	Extended offline	Completed without error	00%	3907	-
#13	Short offline	Completed without error	00%	3890	-
#14	Short offline	Completed without error	00%	3866	-
#15	Short offline	Completed without error	00%	3842	-
#16	Short offline	Completed without error	00%	3818	-
#17	Short offline	Completed without error	00%	3794	-
#18	Short offline	Completed without error	00%	3770	-
#19	Short offline	Completed without error	00%	3746	-

SMART Selective self-test log data structure revision number 1

SPAN	MIN_LBA	MAX_LBA	CURRENT_TEST_STATUS
1	0	0	Not_testing
2	0	0	Not_testing
3	0	0	Not_testing
4	0	0	Not_testing
5	0	0	Not_testing

Selective self-test flags (0x0):
After scanning selected spans, do NOT read-scan remainder of disk.
If Selective self-test is pending on power-up, resume after 0 minute delay.

SCT Status Version: 3
SCT Version (vendor specific): 522 (0x020a)
Device State: Active (0)
Current Temperature: 38 Celsius
Power Cycle Min/Max Temperature: 30/43 Celsius
Lifetime Min/Max Temperature: 19/47 Celsius
Under/Over Temperature Limit Count: 0/125

SCT Temperature History Version: 2
Temperature Sampling Period: 3 minutes
Temperature Logging Interval: 59 minutes
Min/Max recommended Temperature: 0/ 0 Celsius
Min/Max Temperature Limit: 0/ 0 Celsius
Temperature History Size (Index): 128 (115)

Index	Estimated Time	Temperature Celsius
116	2026-08-23 14:20	? -
117	2026-08-23 15:19	28 *****
118	2026-08-23 16:18	? -
119	2026-08-23 17:17	30 *****
120	2026-08-23 18:16	31 *****
121	2026-08-23 19:15	35 *****
122	2026-08-23 20:14	37 *****
123	2026-08-23 21:13	37 *****
124	2026-08-23 22:12	37 *****
125	2026-08-23 23:11	36 *****
126	2026-08-24 00:10	34 *****
127	2026-08-24 01:09	34 *****
0	2026-08-24 02:08	34 *****
1	2026-08-24 03:07	33 *****
2	2026-08-24 04:06	33 *****
3	2026-08-24 05:05	34 *****
...	..(4 skipped).	.. *****
8	2026-08-24 10:00	34 *****
9	2026-08-24 10:59	32 *****
10	2026-08-24 11:58	33 *****
11	2026-08-24 12:57	34 *****
12	2026-08-24 13:56	34 *****
13	2026-08-24 14:55	34 *****
14	2026-08-24 15:54	35 *****
15	2026-08-24 16:53	34 *****
16	2026-08-24 17:52	35 *****
17	2026-08-24 18:51	35 *****
18	2026-08-24 19:50	35 *****

```

19 2026-08-24 20:49 36 *****
... ..( 3 skipped). .. *****
23 2026-08-25 00:45 36 *****
24 2026-08-25 01:44 35 *****
... ..( 7 skipped). .. *****
32 2026-08-25 09:36 35 *****
33 2026-08-25 10:35 34 *****
34 2026-08-25 11:34 35 *****
... ..( 6 skipped). .. *****
41 2026-08-25 18:27 35 *****
42 2026-08-25 19:26 36 *****
... ..( 41 skipped). .. *****
84 2026-08-27 12:44 36 *****
85 2026-08-27 13:43 37 *****
... ..( 3 skipped). .. *****
89 2026-08-27 17:39 37 *****
90 2026-08-27 18:38 36 *****
91 2026-08-27 19:37 37 *****
92 2026-08-27 20:36 37 *****
93 2026-08-27 21:35 36 *****
94 2026-08-27 22:34 36 *****
95 2026-08-27 23:33 36 *****
96 2026-08-28 00:32 38 *****
97 2026-08-28 01:31 43 *****
98 2026-08-28 02:30 43 *****
99 2026-08-28 03:29 42 *****
100 2026-08-28 04:28 41 *****
101 2026-08-28 05:27 41 *****
102 2026-08-28 06:26 41 *****
103 2026-08-28 07:25 40 *****
104 2026-08-28 08:24 41 *****
105 2026-08-28 09:23 40 *****
106 2026-08-28 10:22 40 *****
107 2026-08-28 11:21 40 *****
108 2026-08-28 12:20 41 *****
109 2026-08-28 13:19 42 *****
110 2026-08-28 14:18 42 *****
111 2026-08-28 15:17 38 *****
112 2026-08-28 16:16 36 *****
113 2026-08-28 17:15 36 *****
114 2026-08-28 18:14 36 *****
115 2026-08-28 19:13 37 *****

```

SCT Error Recovery Control:

```

Read: 100 (10.0 seconds)
Write: 100 (10.0 seconds)

```

Device Statistics (GP Log 0x04)

Page	Offset	Size	Value	Flags	Description
0x01	=====	=	=	===	== General Statistics (rev 1) ==
0x01	0x008	4	866	---	Lifetime Power-On Resets
0x01	0x010	4	4363	---	Power-on Hours
0x01	0x018	6	65235131832	---	Logical Sectors Written
0x01	0x020	6	433209885	---	Number of Write Commands
0x01	0x028	6	118064026620	---	Logical Sectors Read
0x01	0x030	6	272891953	---	Number of Read Commands
0x01	0x038	6	-	---	Date and Time TimeStamp
0x03	=====	=	=	===	== Rotating Media Statistics (rev 1) ==
0x03	0x008	4	4315	---	Spindle Motor Power-on Hours
0x03	0x010	4	142	---	Head Flying Hours
0x03	0x018	4	6730	---	Head Load Events
0x03	0x020	4	0	---	Number of Reallocated Logical Sectors
0x03	0x028	4	0	---	Read Recovery Attempts
0x03	0x030	4	0	---	Number of Mechanical Start Failures
0x03	0x038	4	0	---	Number of Realloc. Candidate Logical Sectors
0x03	0x040	4	856	---	Number of High Priority Unload Events
0x04	=====	=	=	===	== General Errors Statistics (rev 1) ==
0x04	0x008	4	0	---	Number of Reported Uncorrectable Errors
0x04	0x010	4	2	---	Resets Between Cmd Acceptance and Completion
0x05	=====	=	=	===	== Temperature Statistics (rev 1) ==
0x05	0x008	1	38	---	Current Temperature
0x05	0x010	1	38	---	Average Short Term Temperature
0x05	0x018	1	32	---	Average Long Term Temperature
0x05	0x020	1	47	---	Highest Temperature
0x05	0x028	1	0	---	Lowest Temperature
0x05	0x030	1	45	---	Highest Average Short Term Temperature
0x05	0x038	1	26	---	Lowest Average Short Term Temperature

```

0x05 0x040 1      32 --- Highest Average Long Term Temperature
0x05 0x048 1      26 --- Lowest Average Long Term Temperature
0x05 0x050 4       0 --- Time in Over-Temperature
0x05 0x058 1      60 --- Specified Maximum Operating Temperature
0x05 0x060 4      30 --- Time in Under-Temperature
0x05 0x068 1       5 --- Specified Minimum Operating Temperature
0x06 ===== =    = === == Transport Statistics (rev 1) ==
0x06 0x008 4     150 --- Number of Hardware Resets
0x06 0x010 4      40 --- Number of ASR Events
0x06 0x018 4       0 --- Number of Interface CRC Errors
0xff ===== =    = === == Vendor Specific Statistics (rev 1) ==
0xff 0x008 7       0 --- Vendor Specific
      ||_ C monitored condition met
      ||_ D supports DSN
      |__ N normalized value

```

Pending Defects log (GP Log 0x0c)
No Defects Logged

SATA Phy Event Counters (GP Log 0x11)

ID	Size	Value	Description
0x000a	2	27	Device-to-host register FISes sent due to a COMRESET
0x0001	2	0	Command failed due to ICRC error
0x0003	2	0	R_ERR response for device-to-host data FIS
0x0004	2	0	R_ERR response for host-to-device data FIS
0x0006	2	0	R_ERR response for device-to-host non-data FIS
0x0007	2	0	R_ERR response for host-to-device non-data FIS

Seagate FARM log (GP Log 0xa6) supported [try: -1 farm]

===== Selbsttestprotokoll =====
smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF READ SMART DATA SECTION ===

SMART Self-test log structure revision number 1

Num	Test_Description	Status	Remaining	LifeTime(hours)	LBA_of_first_error
# 1	Extended offline	Completed without error	00%	4358	-
# 2	Short offline	Completed without error	00%	4343	-
# 3	Short offline	Completed without error	00%	4060	-
# 4	Extended offline	Interrupted (host reset)	60%	4056	-
# 5	Short offline	Interrupted (host reset)	00%	4035	-
# 6	Short offline	Completed without error	00%	4017	-
# 7	Extended offline	Completed without error	00%	3998	-
# 8	Short offline	Completed without error	00%	3969	-
# 9	Short offline	Completed without error	00%	3959	-
#10	Short offline	Completed without error	00%	3938	-
#11	Short offline	Completed without error	00%	3914	-
#12	Extended offline	Completed without error	00%	3907	-
#13	Short offline	Completed without error	00%	3890	-
#14	Short offline	Completed without error	00%	3866	-
#15	Short offline	Completed without error	00%	3842	-
#16	Short offline	Completed without error	00%	3818	-
#17	Short offline	Completed without error	00%	3794	-
#18	Short offline	Completed without error	00%	3770	-
#19	Short offline	Completed without error	00%	3746	-
#20	Extended offline	Completed without error	00%	3740	-
#21	Short offline	Completed without error	00%	3722	-

ZA2DXWL9

```
#####
# Datentraeger /dev/sdg Seriennummer ZA2DXWL9
# Erhoben am 2026-08-28 19:20:57 CEST auf Supermicro X13SAE-F
#####

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF INFORMATION SECTION ===
Model Family:      Seagate Enterprise Capacity 3.5 HDD
Device Model:      ST10000NM0086-2AA101
Serial Number:     ZA2DXWL9
LU WWN Device Id:  5 000c50 0c8198e83
Firmware Version:  SN05
User Capacity:     10,000,831,348,736 bytes [10.0 TB]
Sector Sizes:      512 bytes logical, 4096 bytes physical
Rotation Rate:     7200 rpm
Form Factor:       3.5 inches
Device is:         In smartctl database 7.3/5528
ATA Version is:    ACS-3 T13/2161-D revision 5
SATA Version is:   SATA 3.1, 6.0 Gb/s (current: 6.0 Gb/s)
Local Time is:     Fri Aug 28 19:20:58 2026 CEST
SMART support is:  Available - device has SMART capability.
SMART support is:  Enabled
AAM feature is:    Unavailable
APM feature is:    Unavailable
Rd look-ahead is:  Enabled
Write cache is:    Enabled
DSN feature is:    Unavailable
ATA Security is:   Disabled, NOT FROZEN [SEC1]
Wt Cache Reorder:  Enabled

=== START OF READ SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

General SMART Values:
Offline data collection status:  (0x82) Offline data collection activity
                                   was completed without error.
                                   Auto Offline Data Collection: Enabled.
Self-test execution status:      (   0) The previous self-test routine completed
                                   without error or no self-test has ever
                                   been run.

Total time to complete Offline
data collection:                  ( 567) seconds.
Offline data collection
capabilities:                      (0x7b) SMART execute Offline immediate.
                                   Auto Offline data collection on/off support.
                                   Suspend Offline collection upon new
                                   command.
                                   Offline surface scan supported.
                                   Self-test supported.
                                   Conveyance Self-test supported.
                                   Selective Self-test supported.
SMART capabilities:              (0x0003) Saves SMART data before entering
                                   power-saving mode.
                                   Supports SMART auto save timer.
Error logging capability:        (0x01) Error logging supported.
                                   General Purpose Logging supported.

Short self-test routine
recommended polling time:         (   1) minutes.
Extended self-test routine
recommended polling time:         ( 891) minutes.
Conveyance self-test routine
recommended polling time:         (   2) minutes.
SCT capabilities:                (0x70bd) SCT Status supported.
                                   SCT Error Recovery Control supported.
                                   SCT Feature Control supported.
                                   SCT Data Table supported.

SMART Attributes Data Structure revision number: 10
Vendor Specific SMART Attributes with Thresholds:
```

ID#	ATTRIBUTE_NAME	FLAGS	VALUE	WORST	THRESH	FAIL	RAW_VALUE
1	Raw_Read_Error_Rate	POSR--	083	064	044	-	190331986
3	Spin_Up_Time	P0----	093	091	000	-	0
4	Start_Stop_Count	-0--CK	086	086	020	-	15148
5	Reallocated_Sector_Ct	P0--CK	100	100	010	-	0
7	Seek_Error_Rate	POSR--	090	060	045	-	976236240
9	Power_On_Hours	-0--CK	091	091	000	-	8622
10	Spin_Retry_Count	P0--C-	100	100	097	-	0
12	Power_Cycle_Count	-0--CK	086	086	020	-	15148
184	End-to-End_Error	-0--CK	100	100	099	-	0
187	Reported_Uncorrect	-0--CK	100	100	000	-	0
188	Command_Timeout	-0--CK	100	042	000	-	1 1 122
189	High_Fly_Writes	-0-RCK	001	001	000	-	198
190	Airflow_Temperature_Cel	-0---K	064	054	040	-	36 (Min/Max 29/41)
191	G-Sense_Error_Rate	-0--CK	097	097	000	-	6631
192	Power-Off_Retract_Count	-0--CK	093	093	000	-	15124
193	Load_Cycle_Count	-0--CK	069	069	000	-	62539
194	Temperature_Celsius	-0---K	036	046	000	-	36 (0 19 0 0 0)
195	Hardware_ECC_Recovered	-0-RC-	083	064	000	-	190331986
197	Current_Pending_Sector	-0--C-	100	100	000	-	0
198	Offline_Uncorrectable	----C-	100	100	000	-	0
199	UDMA_CRC_Error_Count	-OSRCK	200	200	000	-	0
200	Pressure_Limit	P0---K	100	100	001	-	0
240	Head_Flying_Hours	-----	100	253	000	-	8313h+09m+30.473s
241	Total_LBAs_Written	-----	100	253	000	-	76934800192
242	Total_LBAs_Read	-----	100	253	000	-	103530074636
		_	K auto-keep				
		_	C event count				
		_	R error rate				
		_ _	S speed/performance				
		_ _ _	O updated online				
		_ _ _	P prefailure warning				

General Purpose Log Directory Version 1

SMART Log Directory Version 1 [multi-sector log support]

Address	Access	R/W	Size	Description
0x00	GPL,SL	R/O	1	Log Directory
0x01	SL	R/O	1	Summary SMART error log
0x02	SL	R/O	5	Comprehensive SMART error log
0x03	GPL	R/O	5	Ext. Comprehensive SMART error log
0x04	GPL	R/O	256	Device Statistics log
0x04	SL	R/O	8	Device Statistics log
0x06	SL	R/O	1	SMART self-test log
0x07	GPL	R/O	1	Extended self-test log
0x08	GPL	R/O	2	Power Conditions log
0x09	SL	R/W	1	Selective self-test log
0x0c	GPL	R/O	2048	Pending Defects log
0x10	GPL	R/O	1	NCQ Command Error log
0x11	GPL	R/O	1	SATA Phy Event Counters log
0x13	GPL	R/O	1	SATA NCQ Send and Receive log
0x15	GPL	R/W	1	Rebuild Assist log
0x21	GPL	R/O	1	Write stream error log
0x22	GPL	R/O	1	Read stream error log
0x24	GPL	R/O	768	Current Device Internal Status Data log
0x30	GPL,SL	R/O	9	IDENTIFY DEVICE data log
0x80-0x9f	GPL,SL	R/W	16	Host vendor specific log
0xa1	GPL,SL	VS	24	Device vendor specific log
0xa2	GPL	VS	16320	Device vendor specific log
0xa6	GPL	VS	192	Device vendor specific log
0xa8-0xa9	GPL,SL	VS	136	Device vendor specific log
0xab	GPL	VS	1	Device vendor specific log
0xad	GPL	VS	16	Device vendor specific log
0xb0	GPL	VS	17208	Device vendor specific log
0xbe-0xbf	GPL	VS	65535	Device vendor specific log
0xc1	GPL,SL	VS	16	Device vendor specific log
0xc3	GPL,SL	VS	56	Device vendor specific log
0xc9	GPL,SL	VS	8	Device vendor specific log
0xca	GPL,SL	VS	16	Device vendor specific log
0xd1	GPL	VS	304	Device vendor specific log
0xd2	GPL	VS	10000	Device vendor specific log
0xd4	GPL	VS	2048	Device vendor specific log
0xe0	GPL,SL	R/W	1	SCT Command/Status
0xe1	GPL,SL	R/W	1	SCT Data Transfer

SMART Extended Comprehensive Error Log Version: 1 (5 sectors)

No Errors Logged

SMART Extended Self-test Log Version: 1 (1 sectors)

Num	Test_Description	Status	Remaining	LifeTime(hours)	LBA_of_first_error
# 1	Extended offline	Completed without error	00%	8616	-
# 2	Short offline	Completed without error	00%	8602	-
# 3	Short offline	Completed without error	00%	8319	-
# 4	Short offline	Completed without error	00%	8295	-
# 5	Short offline	Completed without error	00%	8295	-
# 6	Short offline	Interrupted (host reset)	00%	8294	-
# 7	Short offline	Completed without error	00%	8276	-
# 8	Short offline	Completed without error	00%	8252	-
# 9	Extended offline	Interrupted (host reset)	00%	8244	-
#10	Short offline	Completed without error	00%	8228	-
#11	Short offline	Completed without error	00%	8218	-
#12	Short offline	Completed without error	00%	8197	-
#13	Short offline	Completed without error	00%	8173	-
#14	Extended offline	Completed without error	00%	8164	-
#15	Short offline	Completed without error	00%	8149	-
#16	Short offline	Completed without error	00%	8125	-
#17	Short offline	Completed without error	00%	8101	-
#18	Short offline	Completed without error	00%	8077	-
#19	Short offline	Completed without error	00%	8053	-

SMART Selective self-test log data structure revision number 1

SPAN	MIN_LBA	MAX_LBA	CURRENT_TEST_STATUS
1	0	0	Not_testing
2	0	0	Not_testing
3	0	0	Not_testing
4	0	0	Not_testing
5	0	0	Not_testing

Selective self-test flags (0x0):

After scanning selected spans, do NOT read-scan remainder of disk.
If Selective self-test is pending on power-up, resume after 0 minute delay.

SCT Status Version: 3
SCT Version (vendor specific): 522 (0x020a)
Device State: Active (0)
Current Temperature: 36 Celsius
Power Cycle Min/Max Temperature: 29/41 Celsius
Lifetime Min/Max Temperature: 19/46 Celsius
Under/Over Temperature Limit Count: 0/125

SCT Temperature History Version: 2
Temperature Sampling Period: 3 minutes
Temperature Logging Interval: 59 minutes
Min/Max recommended Temperature: 0/ 0 Celsius
Min/Max Temperature Limit: 0/ 0 Celsius
Temperature History Size (Index): 128 (116)

Index	Estimated Time	Temperature Celsius
117	2026-08-23 14:20	? -
118	2026-08-23 15:19	28 *****
119	2026-08-23 16:18	? -
120	2026-08-23 17:17	29 *****
121	2026-08-23 18:16	30 *****
122	2026-08-23 19:15	33 *****
123	2026-08-23 20:14	35 *****
124	2026-08-23 21:13	35 *****
125	2026-08-23 22:12	35 *****
126	2026-08-23 23:11	34 *****
127	2026-08-24 00:10	33 *****
0	2026-08-24 01:09	33 *****
1	2026-08-24 02:08	32 *****
2	2026-08-24 03:07	31 *****
3	2026-08-24 04:06	32 *****
...	..(5 skipped).	.. *****
9	2026-08-24 10:00	32 *****
10	2026-08-24 10:59	30 *****
11	2026-08-24 11:58	32 *****
12	2026-08-24 12:57	32 *****
13	2026-08-24 13:56	32 *****
14	2026-08-24 14:55	33 *****
...	..(3 skipped).	.. *****
18	2026-08-24 18:51	33 *****
19	2026-08-24 19:50	34 *****
...	..(7 skipped).	.. *****
27	2026-08-25 03:42	34 *****
28	2026-08-25 04:41	33 *****

```
... ..( 9 skipped). .. *****
38 2026-08-25 14:31 33 *****
39 2026-08-25 15:30 34 *****
... ..( 26 skipped). .. *****
66 2026-08-26 18:03 34 *****
67 2026-08-26 19:02 35 *****
68 2026-08-26 20:01 35 *****
69 2026-08-26 21:00 34 *****
70 2026-08-26 21:59 35 *****
... ..( 7 skipped). .. *****
78 2026-08-27 05:51 35 *****
79 2026-08-27 06:50 34 *****
... ..( 3 skipped). .. *****
83 2026-08-27 10:46 34 *****
84 2026-08-27 11:45 35 *****
... ..( 3 skipped). .. *****
88 2026-08-27 15:41 35 *****
89 2026-08-27 16:40 36 *****
90 2026-08-27 17:39 36 *****
91 2026-08-27 18:38 35 *****
92 2026-08-27 19:37 35 *****
93 2026-08-27 20:36 35 *****
94 2026-08-27 21:35 34 *****
95 2026-08-27 22:34 34 *****
96 2026-08-27 23:33 35 *****
97 2026-08-28 00:32 36 *****
98 2026-08-28 01:31 41 *****
99 2026-08-28 02:30 41 *****
100 2026-08-28 03:29 41 *****
101 2026-08-28 04:28 40 *****
102 2026-08-28 05:27 40 *****
103 2026-08-28 06:26 40 *****
104 2026-08-28 07:25 39 *****
... ..( 3 skipped). .. *****
108 2026-08-28 11:21 39 *****
109 2026-08-28 12:20 40 *****
110 2026-08-28 13:19 40 *****
111 2026-08-28 14:18 40 *****
112 2026-08-28 15:17 35 *****
... ..( 3 skipped). .. *****
116 2026-08-28 19:13 35 *****
```

SCT Error Recovery Control:

Read: 100 (10.0 seconds)
Write: 100 (10.0 seconds)

Device Statistics (GP Log 0x04)

Page	Offset	Size	Value	Flags	Description
0x01	=====	=	=	===	== General Statistics (rev 1) ==
0x01	0x008	4	15148	---	Lifetime Power-On Resets
0x01	0x010	4	8622	---	Power-on Hours
0x01	0x018	6	76896736440	---	Logical Sectors Written
0x01	0x020	6	430654784	---	Number of Write Commands
0x01	0x028	6	103480729918	---	Logical Sectors Read
0x01	0x030	6	277104520	---	Number of Read Commands
0x01	0x038	6	-	---	Date and Time TimeStamp
0x03	=====	=	=	===	== Rotating Media Statistics (rev 1) ==
0x03	0x008	4	8523	---	Spindle Motor Power-on Hours
0x03	0x010	4	128	---	Head Flying Hours
0x03	0x018	4	62539	---	Head Load Events
0x03	0x020	4	0	---	Number of Reallocated Logical Sectors
0x03	0x028	4	0	---	Read Recovery Attempts
0x03	0x030	4	0	---	Number of Mechanical Start Failures
0x03	0x038	4	0	---	Number of Realloc. Candidate Logical Sectors
0x03	0x040	4	15119	---	Number of High Priority Unload Events
0x04	=====	=	=	===	== General Errors Statistics (rev 1) ==
0x04	0x008	4	0	---	Number of Reported Uncorrectable Errors
0x04	0x010	4	121	---	Resets Between Cmd Acceptance and Completion
0x05	=====	=	=	===	== Temperature Statistics (rev 1) ==
0x05	0x008	1	36	---	Current Temperature
0x05	0x010	1	37	---	Average Short Term Temperature
0x05	0x018	1	32	---	Average Long Term Temperature
0x05	0x020	1	46	---	Highest Temperature
0x05	0x028	1	0	---	Lowest Temperature
0x05	0x030	1	44	---	Highest Average Short Term Temperature
0x05	0x038	1	26	---	Lowest Average Short Term Temperature
0x05	0x040	1	32	---	Highest Average Long Term Temperature

```

0x05 0x048 1      28 --- Lowest Average Long Term Temperature
0x05 0x050 4      0 --- Time in Over-Temperature
0x05 0x058 1      60 --- Specified Maximum Operating Temperature
0x05 0x060 4      40 --- Time in Under-Temperature
0x05 0x068 1      5 --- Specified Minimum Operating Temperature
0x06 ===== =
0x06 0x008 4      416 --- Number of Hardware Resets
0x06 0x010 4      61 --- Number of ASR Events
0x06 0x018 4      0 --- Number of Interface CRC Errors
0xff ===== =
0xff 0x008 7      0 --- Vendor Specific
                        |||_ C monitored condition met
                        ||__ D supports DSN
                        |___ N normalized value

```

Pending Defects log (GP Log 0x0c)
No Defects Logged

SATA Phy Event Counters (GP Log 0x11)

ID	Size	Value	Description
0x000a	2	27	Device-to-host register FISes sent due to a COMRESET
0x0001	2	0	Command failed due to ICRC error
0x0003	2	0	R_ERR response for device-to-host data FIS
0x0004	2	0	R_ERR response for host-to-device data FIS
0x0006	2	0	R_ERR response for device-to-host non-data FIS
0x0007	2	0	R_ERR response for host-to-device non-data FIS

Seagate FARM log (GP Log 0xa6) supported [try: -l farm]

===== Selbsttestprotokoll =====
smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF READ SMART DATA SECTION ===

SMART Self-test log structure revision number 1

Num	Test_Description	Status	Remaining	LifeTime(hours)	LBA_of_first_error
# 1	Extended offline	Completed without error	00%	8616	-
# 2	Short offline	Completed without error	00%	8602	-
# 3	Short offline	Completed without error	00%	8319	-
# 4	Short offline	Completed without error	00%	8295	-
# 5	Short offline	Completed without error	00%	8295	-
# 6	Short offline	Interrupted (host reset)	00%	8294	-
# 7	Short offline	Completed without error	00%	8276	-
# 8	Short offline	Completed without error	00%	8252	-
# 9	Extended offline	Interrupted (host reset)	00%	8244	-
#10	Short offline	Completed without error	00%	8228	-
#11	Short offline	Completed without error	00%	8218	-
#12	Short offline	Completed without error	00%	8197	-
#13	Short offline	Completed without error	00%	8173	-
#14	Extended offline	Completed without error	00%	8164	-
#15	Short offline	Completed without error	00%	8149	-
#16	Short offline	Completed without error	00%	8125	-
#17	Short offline	Completed without error	00%	8101	-
#18	Short offline	Completed without error	00%	8077	-
#19	Short offline	Completed without error	00%	8053	-
#20	Short offline	Completed without error	00%	8029	-
#21	Short offline	Completed without error	00%	8005	-

ZA2ATQGT

```
#####
# Datentraeger /dev/sd1   Seriennummer ZA2ATQGT
# Erhoben am 2026-08-28 19:21:03 CEST auf Supermicro X13SAE-F
#####

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF INFORMATION SECTION ===
Model Family:      Seagate Enterprise Capacity 3.5 HDD
Device Model:      ST10000NM0086-2AA101
Serial Number:     ZA2ATQGT
LU WWN Device Id:  5 000c50 0b58a82eb
Firmware Version:  SN05
User Capacity:     10,000,831,348,736 bytes [10.0 TB]
Sector Sizes:      512 bytes logical, 4096 bytes physical
Rotation Rate:     7200 rpm
Form Factor:       3.5 inches
Device is:         In smartctl database 7.3/5528
ATA Version is:    ACS-3 T13/2161-D revision 5
SATA Version is:   SATA 3.1, 6.0 Gb/s (current: 6.0 Gb/s)
Local Time is:     Fri Aug 28 19:21:03 2026 CEST
SMART support is:  Available - device has SMART capability.
SMART support is:  Enabled
AAM feature is:    Unavailable
APM feature is:    Unavailable
Rd look-ahead is:  Enabled
Write cache is:    Enabled
DSN feature is:    Unavailable
ATA Security is:   Disabled, NOT FROZEN [SEC1]
Wt Cache Reorder:  Enabled

=== START OF READ SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

General SMART Values:
Offline data collection status:  (0x82) Offline data collection activity
                                   was completed without error.
                                   Auto Offline Data Collection: Enabled.
Self-test execution status:      (   0) The previous self-test routine completed
                                   without error or no self-test has ever
                                   been run.

Total time to complete Offline
data collection:                 ( 567) seconds.
Offline data collection
capabilities:                     (0x7b) SMART execute Offline immediate.
                                   Auto Offline data collection on/off support.
                                   Suspend Offline collection upon new
                                   command.
                                   Offline surface scan supported.
                                   Self-test supported.
                                   Conveyance Self-test supported.
                                   Selective Self-test supported.
SMART capabilities:              (0x0003) Saves SMART data before entering
                                   power-saving mode.
                                   Supports SMART auto save timer.
Error logging capability:        (0x01) Error logging supported.
                                   General Purpose Logging supported.

Short self-test routine
recommended polling time:        (   1) minutes.
Extended self-test routine
recommended polling time:        (  97) minutes.
Conveyance self-test routine
recommended polling time:        (   2) minutes.
SCT capabilities:                (0x70bd) SCT Status supported.
                                   SCT Error Recovery Control supported.
                                   SCT Feature Control supported.
                                   SCT Data Table supported.

SMART Attributes Data Structure revision number: 10
Vendor Specific SMART Attributes with Thresholds:
```

ID#	ATTRIBUTE_NAME	FLAGS	VALUE	WORST	THRESH	FAIL	RAW_VALUE
1	Raw_Read_Error_Rate	POSR--	082	064	044	-	165079806
3	Spin_Up_Time	P0----	093	091	000	-	0
4	Start_Stop_Count	-0--CK	100	100	020	-	256
5	Reallocated_Sector_Ct	P0--CK	100	100	010	-	0
7	Seek_Error_Rate	POSR--	087	060	045	-	508606500
9	Power_On_Hours	-0--CK	090	090	000	-	8888
10	Spin_Retry_Count	P0--C-	100	100	097	-	0
12	Power_Cycle_Count	-0--CK	100	100	020	-	272
184	End-to-End_Error	-0--CK	100	100	099	-	0
187	Reported_Uncorrect	-0--CK	100	100	000	-	0
188	Command_Timeout	-0--CK	100	100	000	-	0 0 1
189	High_Fly_Writes	-0-RCK	025	025	000	-	75
190	Airflow_Temperature_Cel	-0---K	060	048	040	-	40 (Min/Max 32/44)
191	G-Sense_Error_Rate	-0--CK	082	082	000	-	36941
192	Power-Off_Retract_Count	-0--CK	100	100	000	-	549
193	Load_Cycle_Count	-0--CK	100	100	000	-	615
194	Temperature_Celsius	-0---K	040	052	000	-	40 (0 18 0 0 0)
195	Hardware_ECC_Recovered	-0-RC-	082	064	000	-	165079806
197	Current_Pending_Sector	-0--C-	100	100	000	-	0
198	Offline_Uncorrectable	----C-	100	100	000	-	0
199	UDMA_CRC_Error_Count	-QSRCK	200	200	000	-	3
200	Pressure_Limit	P0---K	100	100	001	-	0
240	Head_Flying_Hours	-----	100	253	000	-	8843h+55m+33.265s
241	Total_LBAs_Written	-----	100	253	000	-	71123625545
242	Total_LBAs_Read	-----	100	253	000	-	201078701674
		_	K auto-keep				
		_	C event count				
		_	R error rate				
		_ _	S speed/performance				
		_ _ _	O updated online				
		_ _ _	P prefailure warning				

General Purpose Log Directory Version 1

SMART Log Directory Version 1 [multi-sector log support]

Address	Access	R/W	Size	Description
0x00	GPL,SL	R/O	1	Log Directory
0x01	SL	R/O	1	Summary SMART error log
0x02	SL	R/O	5	Comprehensive SMART error log
0x03	GPL	R/O	5	Ext. Comprehensive SMART error log
0x04	GPL	R/O	256	Device Statistics log
0x04	SL	R/O	8	Device Statistics log
0x06	SL	R/O	1	SMART self-test log
0x07	GPL	R/O	1	Extended self-test log
0x08	GPL	R/O	2	Power Conditions log
0x09	SL	R/W	1	Selective self-test log
0x0c	GPL	R/O	2048	Pending Defects log
0x10	GPL	R/O	1	NCQ Command Error log
0x11	GPL	R/O	1	SATA Phy Event Counters log
0x13	GPL	R/O	1	SATA NCQ Send and Receive log
0x15	GPL	R/W	1	Rebuild Assist log
0x21	GPL	R/O	1	Write stream error log
0x22	GPL	R/O	1	Read stream error log
0x24	GPL	R/O	768	Current Device Internal Status Data log
0x30	GPL,SL	R/O	9	IDENTIFY DEVICE data log
0x80-0x9f	GPL,SL	R/W	16	Host vendor specific log
0xa1	GPL,SL	VS	24	Device vendor specific log
0xa2	GPL	VS	16320	Device vendor specific log
0xa6	GPL	VS	192	Device vendor specific log
0xa8-0xa9	GPL,SL	VS	136	Device vendor specific log
0xab	GPL	VS	1	Device vendor specific log
0xad	GPL	VS	16	Device vendor specific log
0xb0	GPL	VS	17208	Device vendor specific log
0xbe-0xbf	GPL	VS	65535	Device vendor specific log
0xc1	GPL,SL	VS	16	Device vendor specific log
0xc3	GPL,SL	VS	56	Device vendor specific log
0xc9	GPL,SL	VS	8	Device vendor specific log
0xca	GPL,SL	VS	16	Device vendor specific log
0xd1	GPL	VS	304	Device vendor specific log
0xd2	GPL	VS	10000	Device vendor specific log
0xd4	GPL	VS	2048	Device vendor specific log
0xe0	GPL,SL	R/W	1	SCT Command/Status
0xe1	GPL,SL	R/W	1	SCT Data Transfer

SMART Extended Comprehensive Error Log Version: 1 (5 sectors)
No Errors Logged

SMART Extended Self-test Log Version: 1 (1 sectors)

Num	Test_Description	Status	Remaining	LifeTime(hours)	LBA_of_first_error
# 1	Extended offline	Completed without error	00%	8883	-
# 2	Short offline	Completed without error	00%	8868	-
# 3	Short offline	Completed without error	00%	8586	-
# 4	Short offline	Completed without error	00%	8562	-
# 5	Short offline	Completed without error	00%	8561	-
# 6	Short offline	Completed without error	00%	8560	-
# 7	Short offline	Completed without error	00%	8543	-
# 8	Extended offline	Completed without error	00%	8523	-
# 9	Short offline	Completed without error	00%	8495	-
#10	Short offline	Completed without error	00%	8485	-
#11	Short offline	Completed without error	00%	8464	-
#12	Short offline	Completed without error	00%	8440	-
#13	Extended offline	Completed without error	00%	8433	-
#14	Short offline	Completed without error	00%	8416	-
#15	Short offline	Completed without error	00%	8392	-
#16	Short offline	Completed without error	00%	8368	-
#17	Short offline	Completed without error	00%	8344	-
#18	Short offline	Completed without error	00%	8320	-
#19	Short offline	Completed without error	00%	8296	-

SMART Selective self-test log data structure revision number 1

SPAN	MIN_LBA	MAX_LBA	CURRENT_TEST_STATUS
1	0	0	Not_testing
2	0	0	Not_testing
3	0	0	Not_testing
4	0	0	Not_testing
5	0	0	Not_testing

Selective self-test flags (0x0):

After scanning selected spans, do NOT read-scan remainder of disk.
If Selective self-test is pending on power-up, resume after 0 minute delay.

SCT Status Version: 3
SCT Version (vendor specific): 522 (0x020a)
Device State: Active (0)
Current Temperature: 40 Celsius
Power Cycle Min/Max Temperature: 32/44 Celsius
Lifetime Min/Max Temperature: 18/52 Celsius
Under/Over Temperature Limit Count: 0/126

SCT Temperature History Version: 2
Temperature Sampling Period: 3 minutes
Temperature Logging Interval: 59 minutes
Min/Max recommended Temperature: 0/ 0 Celsius
Min/Max Temperature Limit: 0/ 0 Celsius
Temperature History Size (Index): 128 (22)

Index	Estimated Time	Temperature Celsius
23	2026-08-23 14:20	29 *****
24	2026-08-23 15:19	? -
25	2026-08-23 16:18	32 *****
26	2026-08-23 17:17	33 *****
27	2026-08-23 18:16	37 *****
28	2026-08-23 19:15	39 *****
...	..(2 skipped).	.. *****
31	2026-08-23 22:12	39 *****
32	2026-08-23 23:11	36 *****
33	2026-08-24 00:10	37 *****
34	2026-08-24 01:09	37 *****
35	2026-08-24 02:08	36 *****
36	2026-08-24 03:07	35 *****
37	2026-08-24 04:06	37 *****
...	..(4 skipped).	.. *****
42	2026-08-24 09:01	37 *****
43	2026-08-24 10:00	35 *****
44	2026-08-24 10:59	35 *****
45	2026-08-24 11:58	36 *****
46	2026-08-24 12:57	37 *****
...	..(4 skipped).	.. *****
51	2026-08-24 17:52	37 *****
52	2026-08-24 18:51	38 *****
...	..(11 skipped).	.. *****
64	2026-08-25 06:39	38 *****
65	2026-08-25 07:38	37 *****
...	..(3 skipped).	.. *****
69	2026-08-25 11:34	37 *****

```
70 2026-08-25 12:33 38 *****
... ..( 2 skipped). .. *****
73 2026-08-25 15:30 38 *****
74 2026-08-25 16:29 39 *****
75 2026-08-25 17:28 36 *****
76 2026-08-25 18:27 38 *****
77 2026-08-25 19:26 38 *****
78 2026-08-25 20:25 39 *****
... ..( 8 skipped). .. *****
87 2026-08-26 05:16 39 *****
88 2026-08-26 06:15 38 *****
... ..( 6 skipped). .. *****
95 2026-08-26 13:08 38 *****
96 2026-08-26 14:07 39 *****
97 2026-08-26 15:06 39 *****
98 2026-08-26 16:05 39 *****
99 2026-08-26 17:04 37 *****
100 2026-08-26 18:03 38 *****
101 2026-08-26 19:02 39 *****
... ..( 19 skipped). .. *****
121 2026-08-27 14:42 39 *****
122 2026-08-27 15:41 40 *****
123 2026-08-27 16:40 40 *****
124 2026-08-27 17:39 38 *****
125 2026-08-27 18:38 37 *****
126 2026-08-27 19:37 39 *****
127 2026-08-27 20:36 39 *****
0 2026-08-27 21:35 38 *****
1 2026-08-27 22:34 39 *****
2 2026-08-27 23:33 39 *****
3 2026-08-28 00:32 40 *****
4 2026-08-28 01:31 44 *****
5 2026-08-28 02:30 44 *****
6 2026-08-28 03:29 43 *****
7 2026-08-28 04:28 43 *****
8 2026-08-28 05:27 43 *****
9 2026-08-28 06:26 42 *****
10 2026-08-28 07:25 42 *****
11 2026-08-28 08:24 42 *****
12 2026-08-28 09:23 41 *****
13 2026-08-28 10:22 41 *****
14 2026-08-28 11:21 42 *****
15 2026-08-28 12:20 43 *****
16 2026-08-28 13:19 43 *****
17 2026-08-28 14:18 43 *****
18 2026-08-28 15:17 40 *****
19 2026-08-28 16:16 39 *****
20 2026-08-28 17:15 39 *****
21 2026-08-28 18:14 38 *****
22 2026-08-28 19:13 40 *****
```

SCT Error Recovery Control:

```
Read: 100 (10.0 seconds)
Write: 100 (10.0 seconds)
```

Device Statistics (GP Log 0x04)

Page	Offset	Size	Value	Flags	Description
0x01	=====	=	=	===	== General Statistics (rev 1) ==
0x01	0x008	4	272	---	Lifetime Power-On Resets
0x01	0x010	4	8888	---	Power-on Hours
0x01	0x018	6	71123925449	---	Logical Sectors Written
0x01	0x020	6	468964099	---	Number of Write Commands
0x01	0x028	6	200946846599	---	Logical Sectors Read
0x01	0x030	6	658461333	---	Number of Read Commands
0x01	0x038	6	-	---	Date and Time TimeStamp
0x03	=====	=	=	===	== Rotating Media Statistics (rev 1) ==
0x03	0x008	4	8866	---	Spindle Motor Power-on Hours
0x03	0x010	4	791	---	Head Flying Hours
0x03	0x018	4	615	---	Head Load Events
0x03	0x020	4	0	---	Number of Reallocated Logical Sectors
0x03	0x028	4	0	---	Read Recovery Attempts
0x03	0x030	4	0	---	Number of Mechanical Start Failures
0x03	0x038	4	0	---	Number of Realloc. Candidate Logical Sectors
0x03	0x040	4	549	---	Number of High Priority Unload Events
0x04	=====	=	=	===	== General Errors Statistics (rev 1) ==
0x04	0x008	4	0	---	Number of Reported Uncorrectable Errors
0x04	0x010	4	1	---	Resets Between Cmd Acceptance and Completion

```

0x05 ===== =          = === == Temperature Statistics (rev 1) ==
0x05 0x008 1          40 --- Current Temperature
0x05 0x010 1          40 --- Average Short Term Temperature
0x05 0x018 1          35 --- Average Long Term Temperature
0x05 0x020 1          52 --- Highest Temperature
0x05 0x028 1          0 --- Lowest Temperature
0x05 0x030 1          50 --- Highest Average Short Term Temperature
0x05 0x038 1          28 --- Lowest Average Short Term Temperature
0x05 0x040 1          35 --- Highest Average Long Term Temperature
0x05 0x048 1          30 --- Lowest Average Long Term Temperature
0x05 0x050 4          0 --- Time in Over-Temperature
0x05 0x058 1          60 --- Specified Maximum Operating Temperature
0x05 0x060 4          270 --- Time in Under-Temperature
0x05 0x068 1          5 --- Specified Minimum Operating Temperature
0x06 ===== =          = === == Transport Statistics (rev 1) ==
0x06 0x008 4          428 --- Number of Hardware Resets
0x06 0x010 4          140 --- Number of ASR Events
0x06 0x018 4          3 --- Number of Interface CRC Errors
0xff ===== =          = === == Vendor Specific Statistics (rev 1) ==
0xff 0x008 7          0 --- Vendor Specific
      |||_ C monitored condition met
      ||__ D supports DSN
      |___ N normalized value

```

Pending Defects log (GP Log 0x0c)
No Defects Logged

SATA Phy Event Counters (GP Log 0x11)

ID	Size	Value	Description
0x000a	2	27	Device-to-host register FISes sent due to a COMRESET
0x0001	2	0	Command failed due to ICRC error
0x0003	2	0	R_ERR response for device-to-host data FIS
0x0004	2	0	R_ERR response for host-to-device data FIS
0x0006	2	0	R_ERR response for device-to-host non-data FIS
0x0007	2	0	R_ERR response for host-to-device non-data FIS

Seagate FARM log (GP Log 0xa6) supported [try: -l farm]

===== Selbsttestprotokoll =====
smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
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=== START OF READ SMART DATA SECTION ===

SMART Self-test log structure revision number 1

Num	Test_Description	Status	Remaining	LifeTime(hours)	LBA_of_first_error
# 1	Extended offline	Completed without error	00%	8883	-
# 2	Short offline	Completed without error	00%	8868	-
# 3	Short offline	Completed without error	00%	8586	-
# 4	Short offline	Completed without error	00%	8562	-
# 5	Short offline	Completed without error	00%	8561	-
# 6	Short offline	Completed without error	00%	8560	-
# 7	Short offline	Completed without error	00%	8543	-
# 8	Extended offline	Completed without error	00%	8523	-
# 9	Short offline	Completed without error	00%	8495	-
#10	Short offline	Completed without error	00%	8485	-
#11	Short offline	Completed without error	00%	8464	-
#12	Short offline	Completed without error	00%	8440	-
#13	Extended offline	Completed without error	00%	8433	-
#14	Short offline	Completed without error	00%	8416	-
#15	Short offline	Completed without error	00%	8392	-
#16	Short offline	Completed without error	00%	8368	-
#17	Short offline	Completed without error	00%	8344	-
#18	Short offline	Completed without error	00%	8320	-
#19	Short offline	Completed without error	00%	8296	-
#20	Short offline	Completed without error	00%	8272	-
#21	Extended offline	Completed without error	00%	8266	-

ZA2A4DQ1

```
#####
# Datentraeger /dev/sdb Seriennummer ZA2A4DQ1
# Erhoben am 2026-08-28 19:20:52 CEST auf Supermicro X13SAE-F
#####

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF INFORMATION SECTION ===
Model Family:      Seagate Enterprise Capacity 3.5 HDD
Device Model:      ST10000NM0086-2AA101
Serial Number:     ZA2A4DQ1
LU WWN Device Id:  5 000c50 0b3e42ebb
Firmware Version:  SN05
User Capacity:     10,000,831,348,736 bytes [10.0 TB]
Sector Sizes:      512 bytes logical, 4096 bytes physical
Rotation Rate:     7200 rpm
Form Factor:       3.5 inches
Device is:         In smartctl database 7.3/5528
ATA Version is:    ACS-3 T13/2161-D revision 5
SATA Version is:   SATA 3.1, 6.0 Gb/s (current: 6.0 Gb/s)
Local Time is:     Fri Aug 28 19:20:52 2026 CEST
SMART support is:  Available - device has SMART capability.
SMART support is:  Enabled
AAM feature is:    Unavailable
APM feature is:    Unavailable
Rd look-ahead is:  Enabled
Write cache is:    Enabled
DSN feature is:    Unavailable
ATA Security is:   Disabled, NOT FROZEN [SEC1]
Wt Cache Reorder:  Enabled

=== START OF READ SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

General SMART Values:
Offline data collection status:  (0x82) Offline data collection activity
                                   was completed without error.
                                   Auto Offline Data Collection: Enabled.
Self-test execution status:      (   0) The previous self-test routine completed
                                   without error or no self-test has ever
                                   been run.

Total time to complete Offline
data collection:                  ( 567) seconds.
Offline data collection
capabilities:                      (0x7b) SMART execute Offline immediate.
                                   Auto Offline data collection on/off support.
                                   Suspend Offline collection upon new
                                   command.
                                   Offline surface scan supported.
                                   Self-test supported.
                                   Conveyance Self-test supported.
                                   Selective Self-test supported.
SMART capabilities:              (0x0003) Saves SMART data before entering
                                   power-saving mode.
                                   Supports SMART auto save timer.
Error logging capability:        (0x01) Error logging supported.
                                   General Purpose Logging supported.

Short self-test routine
recommended polling time:        (   1) minutes.
Extended self-test routine
recommended polling time:        (  92) minutes.
Conveyance self-test routine
recommended polling time:        (   2) minutes.
SCT capabilities:               (0x70bd) SCT Status supported.
                                   SCT Error Recovery Control supported.
                                   SCT Feature Control supported.
                                   SCT Data Table supported.

SMART Attributes Data Structure revision number: 10
Vendor Specific SMART Attributes with Thresholds:
```

ID#	ATTRIBUTE_NAME	FLAGS	VALUE	WORST	THRESH	FAIL	RAW_VALUE
1	Raw_Read_Error_Rate	POSR--	072	064	044	-	16371246
3	Spin_Up_Time	P0----	093	091	000	-	0
4	Start_Stop_Count	-0--CK	100	100	020	-	266
5	Reallocated_Sector_Ct	P0--CK	100	100	010	-	0
7	Seek_Error_Rate	POSR--	087	060	045	-	466308346
9	Power_On_Hours	-0--CK	057	057	000	-	37805
10	Spin_Retry_Count	P0--C-	100	100	097	-	0
12	Power_Cycle_Count	-0--CK	100	100	020	-	267
184	End-to-End_Error	-0--CK	100	100	099	-	0
187	Reported_Uncorrect	-0--CK	100	100	000	-	0
188	Command_Timeout	-0--CK	100	099	000	-	0 0 2
189	High_Fly_Writes	-0-RCK	001	001	000	-	113
190	Airflow_Temperature_Cel	-0---K	060	049	040	-	40 (Min/Max 32/43)
191	G-Sense_Error_Rate	-0--CK	084	084	000	-	33637
192	Power-Off_Retract_Count	-0--CK	100	100	000	-	1430
193	Load_Cycle_Count	-0--CK	100	100	000	-	1785
194	Temperature_Celsius	-0---K	040	051	000	-	40 (0 19 0 0 0)
195	Hardware_ECC_Recovered	-0-RC-	072	064	000	-	16371246
197	Current_Pending_Sector	-0--C-	100	100	000	-	0
198	Offline_Uncorrectable	----C-	100	100	000	-	0
199	UDMA_CRC_Error_Count	-QSRCK	200	200	000	-	0
200	Pressure_Limit	P0---K	100	100	001	-	0
240	Head_Flying_Hours	-----	100	253	000	-	37603h+49m+27.556s
241	Total_LBAs_Written	-----	100	253	000	-	157654103667
242	Total_LBAs_Read	-----	100	253	000	-	508369580183

|||||_ K auto-keep
 |||||_ C event count
 ||||_ R error rate
 |||_ S speed/performance
 ||_ O updated online
 |_ P prefailure warning

General Purpose Log Directory Version 1

SMART Log Directory Version 1 [multi-sector log support]

Address	Access	R/W	Size	Description
0x00	GPL,SL	R/O	1	Log Directory
0x01	SL	R/O	1	Summary SMART error log
0x02	SL	R/O	5	Comprehensive SMART error log
0x03	GPL	R/O	5	Ext. Comprehensive SMART error log
0x04	GPL	R/O	256	Device Statistics log
0x04	SL	R/O	8	Device Statistics log
0x06	SL	R/O	1	SMART self-test log
0x07	GPL	R/O	1	Extended self-test log
0x08	GPL	R/O	2	Power Conditions log
0x09	SL	R/W	1	Selective self-test log
0x0c	GPL	R/O	2048	Pending Defects log
0x10	GPL	R/O	1	NCQ Command Error log
0x11	GPL	R/O	1	SATA Phy Event Counters log
0x13	GPL	R/O	1	SATA NCQ Send and Receive log
0x15	GPL	R/W	1	Rebuild Assist log
0x21	GPL	R/O	1	Write stream error log
0x22	GPL	R/O	1	Read stream error log
0x24	GPL	R/O	768	Current Device Internal Status Data log
0x30	GPL,SL	R/O	9	IDENTIFY DEVICE data log
0x80-0x9f	GPL,SL	R/W	16	Host vendor specific log
0xa1	GPL,SL	VS	24	Device vendor specific log
0xa2	GPL	VS	16320	Device vendor specific log
0xa6	GPL	VS	192	Device vendor specific log
0xa8-0xa9	GPL,SL	VS	136	Device vendor specific log
0xab	GPL	VS	1	Device vendor specific log
0xad	GPL	VS	16	Device vendor specific log
0xb0	GPL	VS	17208	Device vendor specific log
0xbe-0xbf	GPL	VS	65535	Device vendor specific log
0xc1	GPL,SL	VS	16	Device vendor specific log
0xc3	GPL,SL	VS	56	Device vendor specific log
0xc9	GPL,SL	VS	8	Device vendor specific log
0xca	GPL,SL	VS	16	Device vendor specific log
0xd1	GPL	VS	304	Device vendor specific log
0xd2	GPL	VS	10000	Device vendor specific log
0xd4	GPL	VS	2048	Device vendor specific log
0xe0	GPL,SL	R/W	1	SCT Command/Status
0xe1	GPL,SL	R/W	1	SCT Data Transfer

SMART Extended Comprehensive Error Log Version: 1 (5 sectors)

No Errors Logged

SMART Extended Self-test Log Version: 1 (1 sectors)

Num	Test_Description	Status	Remaining	LifeTime(hours)	LBA_of_first_error
# 1	Extended offline	Completed without error	00%	37800	-
# 2	Short offline	Completed without error	00%	37785	-
# 3	Short offline	Completed without error	00%	37503	-
# 4	Short offline	Completed without error	00%	37479	-
# 5	Short offline	Completed without error	00%	37478	-
# 6	Short offline	Completed without error	00%	37477	-
# 7	Short offline	Completed without error	00%	37460	-
# 8	Extended offline	Completed without error	00%	37441	-
# 9	Short offline	Completed without error	00%	37412	-
#10	Short offline	Completed without error	00%	37402	-
#11	Short offline	Completed without error	00%	37381	-
#12	Short offline	Completed without error	00%	37357	-
#13	Extended offline	Completed without error	00%	37350	-
#14	Short offline	Completed without error	00%	37333	-
#15	Short offline	Completed without error	00%	37309	-
#16	Short offline	Completed without error	00%	37285	-
#17	Short offline	Completed without error	00%	37261	-
#18	Short offline	Completed without error	00%	37237	-
#19	Short offline	Completed without error	00%	37213	-

SMART Selective self-test log data structure revision number 1

SPAN	MIN_LBA	MAX_LBA	CURRENT_TEST_STATUS
1	0	0	Not_testing
2	0	0	Not_testing
3	0	0	Not_testing
4	0	0	Not_testing
5	0	0	Not_testing

Selective self-test flags (0x0):

After scanning selected spans, do NOT read-scan remainder of disk.
If Selective self-test is pending on power-up, resume after 0 minute delay.

SCT Status Version: 3
SCT Version (vendor specific): 522 (0x020a)
Device State: Active (0)
Current Temperature: 40 Celsius
Power Cycle Min/Max Temperature: 32/43 Celsius
Lifetime Min/Max Temperature: 19/51 Celsius
Under/Over Temperature Limit Count: 0/127

SCT Temperature History Version: 2
Temperature Sampling Period: 3 minutes
Temperature Logging Interval: 59 minutes
Min/Max recommended Temperature: 0/ 0 Celsius
Min/Max Temperature Limit: 0/ 0 Celsius
Temperature History Size (Index): 128 (98)

Index	Estimated Time	Temperature Celsius
99	2026-08-23 14:20	? -
100	2026-08-23 15:19	32 *****
101	2026-08-23 16:18	33 *****
102	2026-08-23 17:17	37 *****
103	2026-08-23 18:16	38 *****
104	2026-08-23 19:15	39 *****
105	2026-08-23 20:14	39 *****
106	2026-08-23 21:13	38 *****
107	2026-08-23 22:12	36 *****
...	..(2 skipped).	.. *****
110	2026-08-24 01:09	36 *****
111	2026-08-24 02:08	35 *****
112	2026-08-24 03:07	36 *****
...	..(4 skipped).	.. *****
117	2026-08-24 08:02	36 *****
118	2026-08-24 09:01	35 *****
119	2026-08-24 10:00	35 *****
120	2026-08-24 10:59	36 *****
121	2026-08-24 11:58	36 *****
122	2026-08-24 12:57	37 *****
...	..(2 skipped).	.. *****
125	2026-08-24 15:54	37 *****
126	2026-08-24 16:53	35 *****
127	2026-08-24 17:52	37 *****
0	2026-08-24 18:51	38 *****
...	..(6 skipped).	.. *****
7	2026-08-25 01:44	38 *****
8	2026-08-25 02:43	37 *****

```

... ..( 10 skipped). .. *****
19 2026-08-25 13:32 37 *****
20 2026-08-25 14:31 38 *****
21 2026-08-25 15:30 38 *****
22 2026-08-25 16:29 37 *****
23 2026-08-25 17:28 38 *****
24 2026-08-25 18:27 36 *****
25 2026-08-25 19:26 38 *****
... ..( 22 skipped). .. *****
48 2026-08-26 18:03 38 *****
49 2026-08-26 19:02 39 *****
50 2026-08-26 20:01 37 *****
51 2026-08-26 21:00 39 *****
... ..( 6 skipped). .. *****
58 2026-08-27 03:53 39 *****
59 2026-08-27 04:52 38 *****
... ..( 6 skipped). .. *****
66 2026-08-27 11:45 38 *****
67 2026-08-27 12:44 39 *****
... ..( 4 skipped). .. *****
72 2026-08-27 17:39 39 *****
73 2026-08-27 18:38 38 *****
74 2026-08-27 19:37 39 *****
75 2026-08-27 20:36 37 *****
76 2026-08-27 21:35 37 *****
77 2026-08-27 22:34 38 *****
78 2026-08-27 23:33 39 *****
79 2026-08-28 00:32 41 *****
80 2026-08-28 01:31 43 *****
81 2026-08-28 02:30 43 *****
82 2026-08-28 03:29 43 *****
83 2026-08-28 04:28 42 *****
84 2026-08-28 05:27 42 *****
85 2026-08-28 06:26 42 *****
86 2026-08-28 07:25 41 *****
... ..( 3 skipped). .. *****
90 2026-08-28 11:21 41 *****
91 2026-08-28 12:20 42 *****
92 2026-08-28 13:19 42 *****
93 2026-08-28 14:18 42 *****
94 2026-08-28 15:17 39 *****
... ..( 2 skipped). .. *****
97 2026-08-28 18:14 39 *****
98 2026-08-28 19:13 40 *****

```

SCT Error Recovery Control:

Read: 100 (10.0 seconds)

Write: 100 (10.0 seconds)

Device Statistics (GP Log 0x04)

Page	Offset	Size	Value	Flags	Description
0x01	=====	=	=	===	== General Statistics (rev 1) ==
0x01	0x008	4	267	---	Lifetime Power-On Resets
0x01	0x010	4	37805	---	Power-on Hours
0x01	0x018	6	157653882513	---	Logical Sectors Written
0x01	0x020	6	1040471869	---	Number of Write Commands
0x01	0x028	6	508368476809	---	Logical Sectors Read
0x01	0x030	6	1379234497	---	Number of Read Commands
0x01	0x038	6	-	---	Date and Time TimeStamp
0x03	=====	=	=	===	== Rotating Media Statistics (rev 1) ==
0x03	0x008	4	37786	---	Spindle Motor Power-on Hours
0x03	0x010	4	1531	---	Head Flying Hours
0x03	0x018	4	1785	---	Head Load Events
0x03	0x020	4	0	---	Number of Reallocated Logical Sectors
0x03	0x028	4	0	---	Read Recovery Attempts
0x03	0x030	4	0	---	Number of Mechanical Start Failures
0x03	0x038	4	0	---	Number of Realloc. Candidate Logical Sectors
0x03	0x040	4	1430	---	Number of High Priority Unload Events
0x04	=====	=	=	===	== General Errors Statistics (rev 1) ==
0x04	0x008	4	0	---	Number of Reported Uncorrectable Errors
0x04	0x010	4	2	---	Resets Between Cmd Acceptance and Completion
0x05	=====	=	=	===	== Temperature Statistics (rev 1) ==
0x05	0x008	1	40	---	Current Temperature
0x05	0x010	1	40	---	Average Short Term Temperature
0x05	0x018	1	35	---	Average Long Term Temperature
0x05	0x020	1	51	---	Highest Temperature
0x05	0x028	1	0	---	Lowest Temperature

```

0x05 0x030 1          49 --- Highest Average Short Term Temperature
0x05 0x038 1          28 --- Lowest Average Short Term Temperature
0x05 0x040 1          37 --- Highest Average Long Term Temperature
0x05 0x048 1          30 --- Lowest Average Long Term Temperature
0x05 0x050 4           0 --- Time in Over-Temperature
0x05 0x058 1          60 --- Specified Maximum Operating Temperature
0x05 0x060 4         200 --- Time in Under-Temperature
0x05 0x068 1           5 --- Specified Minimum Operating Temperature
0x06 ===== =       = === == Transport Statistics (rev 1) ==
0x06 0x008 4         177 --- Number of Hardware Resets
0x06 0x010 4          48 --- Number of ASR Events
0x06 0x018 4           0 --- Number of Interface CRC Errors
0xff ===== =       = === == Vendor Specific Statistics (rev 1) ==
0xff 0x008 7           0 --- Vendor Specific
    ||_ C monitored condition met
    ||_ D supports DSN
    |__ N normalized value

```

Pending Defects log (GP Log 0x0c)

No Defects Logged

SATA Phy Event Counters (GP Log 0x11)

ID	Size	Value	Description
0x000a	2	27	Device-to-host register FISes sent due to a COMRESET
0x0001	2	0	Command failed due to ICRC error
0x0003	2	0	R_ERR response for device-to-host data FIS
0x0004	2	0	R_ERR response for host-to-device data FIS
0x0006	2	0	R_ERR response for device-to-host non-data FIS
0x0007	2	0	R_ERR response for host-to-device non-data FIS

Seagate FARM log (GP Log 0xa6) supported [try: -1 farm]

===== Selbsttestprotokoll =====

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)

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=== START OF READ SMART DATA SECTION ===

SMART Self-test log structure revision number 1

Num	Test_Description	Status	Remaining	LifeTime(hours)	LBA_of_first_error
# 1	Extended offline	Completed without error	00%	37800	-
# 2	Short offline	Completed without error	00%	37785	-
# 3	Short offline	Completed without error	00%	37503	-
# 4	Short offline	Completed without error	00%	37479	-
# 5	Short offline	Completed without error	00%	37478	-
# 6	Short offline	Completed without error	00%	37477	-
# 7	Short offline	Completed without error	00%	37460	-
# 8	Extended offline	Completed without error	00%	37441	-
# 9	Short offline	Completed without error	00%	37412	-
#10	Short offline	Completed without error	00%	37402	-
#11	Short offline	Completed without error	00%	37381	-
#12	Short offline	Completed without error	00%	37357	-
#13	Extended offline	Completed without error	00%	37350	-
#14	Short offline	Completed without error	00%	37333	-
#15	Short offline	Completed without error	00%	37309	-
#16	Short offline	Completed without error	00%	37285	-
#17	Short offline	Completed without error	00%	37261	-
#18	Short offline	Completed without error	00%	37237	-
#19	Short offline	Completed without error	00%	37213	-
#20	Short offline	Completed without error	00%	37189	-
#21	Extended offline	Completed without error	00%	37183	-

BTWL341407UP800RGN

```
#####
# Datentraeger /dev/sdj   Seriennummer BTWL341407UP800RGN
# Erhoben am 2026-08-28 19:21:01 CEST auf Supermicro X13SAE-F
#####

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF INFORMATION SECTION ===
Device Model:          VK0800GDJYA
Serial Number:         BTWL341407UP800RGN
LU WWN Device Id:      5 5cd2e4 04b4e59d8
Firmware Version:      4IWHVPG0
User Capacity:         800,166,076,416 bytes [800 GB]
Sector Size:           512 bytes logical/physical
Rotation Rate:         Solid State Device
Form Factor:           2.5 inches
TRIM Command:          Available, deterministic, zeroed
Device is:             Not in smartctl database 7.3/5528
ATA Version is:        ATA8-ACS T13/1699-D revision 4
SATA Version is:       SATA 2.6, 6.0 Gb/s (current: 6.0 Gb/s)
Local Time is:         Fri Aug 28 19:21:01 2026 CEST
SMART support is:      Available - device has SMART capability.
SMART support is:      Enabled
AAM feature is:        Unavailable
APM feature is:        Unavailable
Rd look-ahead is:      Enabled
Write cache is:        Disabled
DSN feature is:        Unavailable
ATA Security is:       Unavailable
Wt Cache Reorder:      Enabled

=== START OF READ SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

General SMART Values:
Offline data collection status:  (0x02) Offline data collection activity
                                   was completed without error.
                                   Auto Offline Data Collection: Disabled.
Self-test execution status:      (   0) The previous self-test routine completed
                                   without error or no self-test has ever
                                   been run.
Total time to complete Offline
data collection:                 (   2) seconds.
Offline data collection
capabilities:                     (0x79) SMART execute Offline immediate.
                                   No Auto Offline data collection support.
                                   Suspend Offline collection upon new
                                   command.
                                   Offline surface scan supported.
                                   Self-test supported.
                                   Conveyance Self-test supported.
                                   Selective Self-test supported.
SMART capabilities:              (0x0003) Saves SMART data before entering
                                   power-saving mode.
                                   Supports SMART auto save timer.
Error logging capability:        (0x01) Error logging supported.
                                   General Purpose Logging supported.
Short self-test routine
recommended polling time:        (   1) minutes.
Extended self-test routine
recommended polling time:        (   2) minutes.
Conveyance self-test routine
recommended polling time:        (   2) minutes.
SCT capabilities:               (0x0039) SCT Status supported.
                                   SCT Error Recovery Control supported.
                                   SCT Feature Control supported.
                                   SCT Data Table supported.

SMART Attributes Data Structure revision number: 1
Vendor Specific SMART Attributes with Thresholds:
```

ID#	ATTRIBUTE_NAME	FLAGS	VALUE	WORST	THRESH	FAIL	RAW_VALUE
1	Raw_Read_Error_Rate	POSR--	130	130	039	-	20609124
5	Reallocated_Sector_Ct	PO--CK	097	097	001	-	0
9	Power_On_Hours	-O--CK	100	100	000	-	92518
12	Power_Cycle_Count	-O--CK	100	100	000	-	172
171	Unknown_Attribute	-O--CK	100	100	000	-	0
172	Unknown_Attribute	-O--CK	100	100	000	-	0
173	Unknown_Attribute	PO--CK	092	092	001	-	0
174	Unknown_Attribute	-O--CK	100	100	000	-	161
175	Program_Fail_Count_Chip	PO--CK	100	100	010	-	2302591238786
180	Unused_Rsvd_Blkc_Cnt_Tot	PO-RCK	130	130	039	-	76314276
184	End-to-End_Error	PO--CK	100	100	000	-	0
187	Reported_Uncorrect	-O--CK	100	100	000	-	0
188	Command_Timeout	-O--CK	100	100	000	-	0
190	Airflow_Temperature_Cel	-O---K	074	073	000	-	26 (Min/Max 20/27)
194	Temperature_Celsius	-O---K	100	100	000	-	26
195	Hardware_ECC_Recovered	-O-RC-	100	100	000	-	4638133
196	Reallocated_Event_Count	PO--CK	100	100	010	-	0
197	Current_Pending_Sector	-O--C-	100	100	000	-	0
198	Offline_Uncorrectable	---C-	100	100	000	-	0
199	UDMA_CRC_Error_Count	-OSRCK	100	100	000	-	0
226	Unknown_SSD_Attribute	-O--CK	100	100	000	-	0
227	Unknown_SSD_Attribute	-O--CK	100	100	000	-	27
228	Power-off_Retract_Count	-O--CK	100	100	000	-	7458

|||||_ K auto-keep
 |||||_ C event count
 |||||_ R error rate
 |||____ S speed/performance
 ||____ O updated online
 |____ P prefailure warning

General Purpose Log Directory Version 1

SMART Log Directory Version 1 [multi-sector log support]

Address	Access	R/W	Size	Description
0x00	GPL,SL	R/O	1	Log Directory
0x01	GPL,SL	R/O	1	Summary SMART error log
0x02	GPL,SL	R/O	8	Comprehensive SMART error log
0x03	GPL,SL	R/O	20	Ext. Comprehensive SMART error log
0x04	GPL,SL	R/O	8	Device Statistics log
0x06	GPL,SL	R/O	1	SMART self-test log
0x07	GPL,SL	R/O	2	Extended self-test log
0x09	GPL,SL	R/W	1	Selective self-test log
0x10	GPL,SL	R/O	1	NCQ Command Error log
0x11	GPL,SL	R/O	1	SATA Phy Event Counters log
0x80-0x9f	GPL,SL	R/W	16	Host vendor specific log
0xb0	GPL,SL	VS	1	Device vendor specific log
0xb1	GPL,SL	VS	33	Device vendor specific log
0xb6	GPL,SL	VS	127	Device vendor specific log
0xd0	GPL,SL	VS	1	Device vendor specific log
0xd1	GPL	VS	13056	Device vendor specific log
0xdf	GPL	VS	1152	Device vendor specific log
0xdf	SL	VS	128	Device vendor specific log
0xe0	GPL,SL	R/W	1	SCT Command/Status
0xe1	GPL,SL	R/W	1	SCT Data Transfer

SMART Extended Comprehensive Error Log Version: 1 (20 sectors)

Device Error Count: 4

CR = Command Register
 FEATR = Features Register
 COUNT = Count (was: Sector Count) Register
 LBA_48 = Upper bytes of LBA High/Mid/Low Registers] ATA-8
 LH = LBA High (was: Cylinder High) Register] LBA
 LM = LBA Mid (was: Cylinder Low) Register] Register
 LL = LBA Low (was: Sector Number) Register]
 DV = Device (was: Device/Head) Register
 DC = Device Control Register
 ER = Error register
 ST = Status register

Powered_Up_Time is measured from power on, and printed as

DDd+hh:mm:ss.sss where DD=days, hh=hours, mm=minutes,
 SS=sec, and sss=millisec. It "wraps" after 49.710 days.

Error 4 [3] occurred at disk power-on lifetime: 5962 hours (248 days + 10 hours)
 When the command that caused the error occurred, the device was active or idle.

After command completion occurred, registers were:

ER -- ST COUNT LBA_48 LH LM LL DV DC

```
-- -- -- == -- == == -- -- -- --
00 -- 51 02 00 00 5b ce 00 cc 00 40 00 Error: at LBA = 0x5bce00cc00 = 394298182656
```

Commands leading to the command that caused the error were:

CR	FEATR	COUNT	LBA_48	LH	LM	LL	DV	DC	Powered_Up_Time	Command/Feature_Name
60	00	00	02 00 00 5b ce 00 cc 00 40 00	11d+22:03:16.591	READ FPDMA QUEUED					
60	00	00	02 00 00 5b d0 00 cc 00 40 00	11d+22:03:16.587	READ FPDMA QUEUED					
60	00	00	02 00 00 5b d2 00 cc 00 40 00	11d+22:03:16.583	READ FPDMA QUEUED					
60	00	00	02 00 00 5b d4 00 cc 00 40 00	11d+22:03:16.579	READ FPDMA QUEUED					
60	00	00	02 00 00 5b d6 00 cc 00 40 00	11d+22:03:16.575	READ FPDMA QUEUED					

Error 3 [2] occurred at disk power-on lifetime: 5098 hours (212 days + 10 hours)
When the command that caused the error occurred, the device was active or idle.

After command completion occurred, registers were:

```
ER -- ST COUNT LBA_48 LH LM LL DV DC
-- -- -- == -- == == -- -- -- --
00 -- 51 02 00 00 03 84 00 f7 00 40 00 Error: at LBA = 0x38400f700 = 15099557632
```

Commands leading to the command that caused the error were:

CR	FEATR	COUNT	LBA_48	LH	LM	LL	DV	DC	Powered_Up_Time	Command/Feature_Name
60	00	00	02 00 00 03 84 00 f7 00 40 00	00:10:12.817	READ FPDMA QUEUED					
60	00	00	02 00 00 03 82 00 f7 00 40 00	00:10:12.813	READ FPDMA QUEUED					
60	00	00	02 00 00 03 80 00 f7 00 40 00	00:10:12.809	READ FPDMA QUEUED					
60	00	00	02 00 00 03 7e 00 f7 00 40 00	00:10:12.805	READ FPDMA QUEUED					
60	00	00	02 00 00 03 7c 00 f7 00 40 00	00:10:12.802	READ FPDMA QUEUED					

Error 2 [1] occurred at disk power-on lifetime: 2616 hours (109 days + 0 hours)
When the command that caused the error occurred, the device was active or idle.

After command completion occurred, registers were:

```
ER -- ST COUNT LBA_48 LH LM LL DV DC
-- -- -- == -- == == -- -- -- --
00 -- 51 02 00 00 06 2e 00 9f 00 40 00 Error: at LBA = 0x62e009f00 = 26541596416
```

Commands leading to the command that caused the error were:

CR	FEATR	COUNT	LBA_48	LH	LM	LL	DV	DC	Powered_Up_Time	Command/Feature_Name
60	00	00	02 00 00 06 2e 00 9f 00 40 00	00:23:48.437	READ FPDMA QUEUED					
60	00	00	02 00 00 06 2c 00 9f 00 40 00	00:23:48.432	READ FPDMA QUEUED					
60	00	00	02 00 00 06 2a 00 9f 00 40 00	00:23:48.428	READ FPDMA QUEUED					
60	00	00	02 00 00 06 28 00 9f 00 40 00	00:23:48.424	READ FPDMA QUEUED					
60	00	00	02 00 00 06 26 00 9f 00 40 00	00:23:48.420	READ FPDMA QUEUED					

Error 1 [0] occurred at disk power-on lifetime: 0 hours (0 days + 0 hours)
When the command that caused the error occurred, the device was active or idle.

After command completion occurred, registers were:

```
ER -- ST COUNT LBA_48 LH LM LL DV DC
-- -- -- == -- == == -- -- -- --
00 -- 51 02 00 00 0b 6c 00 e6 00 40 00 Error: at LBA = 0xb6c00e600 = 49056638464
```

Commands leading to the command that caused the error were:

CR	FEATR	COUNT	LBA_48	LH	LM	LL	DV	DC	Powered_Up_Time	Command/Feature_Name
60	00	00	02 00 00 0b 6c 00 e6 00 40 00	00:02:11.317	READ FPDMA QUEUED					
60	00	00	02 00 00 0b 6a 00 e6 00 40 00	00:02:11.316	READ FPDMA QUEUED					
60	00	00	02 00 00 0b 68 00 e6 00 40 00	00:02:11.314	READ FPDMA QUEUED					
60	00	00	02 00 00 0b 66 00 e6 00 40 00	00:02:11.313	READ FPDMA QUEUED					
60	00	00	02 00 00 0b 64 00 e6 00 40 00	00:02:11.311	READ FPDMA QUEUED					

SMART Extended Self-test Log Version: 1 (2 sectors)

Num	Test_Description	Status	Remaining	LifeTime(hours)	LBA_of_first_error
# 1	Short offline	Completed without error	00%	26962	-
# 2	Extended offline	Completed without error	00%	26767	-

SMART Selective self-test log data structure revision number 1

SPAN	MIN_LBA	MAX_LBA	CURRENT_TEST_STATUS
1	0	0	Not_testing
2	0	0	Not_testing
3	0	0	Not_testing
4	0	0	Not_testing
5	0	0	Not_testing

Selective self-test flags (0x0):

After scanning selected spans, do NOT read-scan remainder of disk.

If Selective self-test is pending on power-up, resume after 0 minute delay.

SCT Status Version: 3
SCT Version (vendor specific): 1 (0x0001)
Device State: Active (0)
Current Temperature: 26 Celsius
Power Cycle Min/Max Temperature: 20/27 Celsius
Lifetime Min/Max Temperature: 11/46 Celsius
Specified Max Operating Temperature: 55 Celsius
Under/Over Temperature Limit Count: 0/0
Vendor specific:
00 00 5f 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

SCT Temperature History Version: 2
Temperature Sampling Period: 1 minute
Temperature Logging Interval: 1 minute
Min/Max recommended Temperature: 0/55 Celsius
Min/Max Temperature Limit: 0/55 Celsius
Temperature History Size (Index): 478 (338)

Index	Estimated Time	Temperature Celsius
339	2026-08-28 11:24	25 *****
...	..(105 skipped).	.. *****
445	2026-08-28 13:10	25 *****
446	2026-08-28 13:11	26 *****
447	2026-08-28 13:12	25 *****
448	2026-08-28 13:13	26 *****
...	..(40 skipped).	.. *****
11	2026-08-28 13:54	26 *****
12	2026-08-28 13:55	25 *****
...	..(169 skipped).	.. *****
182	2026-08-28 16:45	25 *****
183	2026-08-28 16:46	24 *****
184	2026-08-28 16:47	24 *****
185	2026-08-28 16:48	25 *****
186	2026-08-28 16:49	24 *****
187	2026-08-28 16:50	25 *****
...	..(71 skipped).	.. *****
259	2026-08-28 18:02	25 *****
260	2026-08-28 18:03	26 *****
...	..(77 skipped).	.. *****
338	2026-08-28 19:21	26 *****

SCT Error Recovery Control:
Read: Disabled
Write: Disabled

Device Statistics (GP Log 0x04)

Page	Offset	Size	Value	Flags	Description
0x01	=====	=	=	===	== General Statistics (rev 2) ==
0x01	0x008	4	172	---	Lifetime Power-On Resets
0x01	0x018	6	82038541834	---	Logical Sectors Written
0x01	0x020	6	3993379	---	Number of Write Commands
0x01	0x028	6	1757420480646	---	Logical Sectors Read
0x01	0x030	6	166948	---	Number of Read Commands
0x04	=====	=	=	===	== General Errors Statistics (rev 1) ==
0x04	0x008	4	0	---	Number of Reported Uncorrectable Errors
0x04	0x010	4	0	---	Resets Between Cmd Acceptance and Completion
0x05	=====	=	=	===	== Temperature Statistics (rev 1) ==
0x05	0x008	1	26	---	Current Temperature
0x05	0x010	1	24	---	Average Short Term Temperature
0x05	0x018	1	18	---	Average Long Term Temperature
0x05	0x020	1	46	---	Highest Temperature
0x05	0x028	1	11	---	Lowest Temperature
0x05	0x030	1	29	---	Highest Average Short Term Temperature
0x05	0x038	1	11	---	Lowest Average Short Term Temperature
0x05	0x040	1	27	---	Highest Average Long Term Temperature
0x05	0x048	1	14	---	Lowest Average Long Term Temperature
0x05	0x050	4	0	---	Time in Over-Temperature
0x05	0x058	1	55	---	Specified Maximum Operating Temperature
0x05	0x060	4	0	---	Time in Under-Temperature
0x05	0x068	1	0	---	Specified Minimum Operating Temperature
0x06	=====	=	=	===	== Transport Statistics (rev 1) ==
0x06	0x008	4	106738	---	Number of Hardware Resets
0x06	0x010	4	489	---	Number of ASR Events
0x06	0x018	4	0	---	Number of Interface CRC Errors

```
0x07 ===== =
0x07 0x008 1      8 --- Percentage Used Endurance Indicator
                        |||_ C monitored condition met
                        ||__ D supports DSN
                        |___ N normalized value

Pending Defects log (GP Log 0x0c) not supported

SATA Phy Event Counters (GP Log 0x11)
ID      Size      Value Description
0x0001  4          0 Command failed due to ICRC error
0x0003  4          0 R_ERR response for device-to-host data FIS
0x0004  4          0 R_ERR response for host-to-device data FIS
0x0006  4          0 R_ERR response for device-to-host non-data FIS
0x000a  4         27 Device-to-host register FISes sent due to a COMRESET
0x000b  4          0 CRC errors within host-to-device FIS
0x000d  4          0 Non-CRC errors within host-to-device FIS

===== Selbsttestprotokoll =====
smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF READ SMART DATA SECTION ===
SMART Self-test log structure revision number 1
Num  Test_Description      Status              Remaining  LifeTime(hours)  LBA_of_first_error
# 1  Short offline          Completed without error    00%          26962             -
# 2  Extended offline        Completed without error    00%          26767             -
```

BTWL3414088B800RGN

```
#####
# Datentraeger /dev/sde   Seriennummer BTWL3414088B800RGN
# Erhoben am 2026-08-28 19:20:55 CEST auf Supermicro X13SAE-F
#####

smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF INFORMATION SECTION ===
Device Model:          VK0800GDJYA
Serial Number:         BTWL3414088B800RGN
LU WWN Device Id: 5 5cd2e4 04b4e5b39
Firmware Version: 4IWWHPG0
User Capacity:         800,166,076,416 bytes [800 GB]
Sector Size:           512 bytes logical/physical
Rotation Rate:         Solid State Device
Form Factor:           2.5 inches
TRIM Command:          Available, deterministic, zeroed
Device is:             Not in smartctl database 7.3/5528
ATA Version is:        ATA8-ACS T13/1699-D revision 4
SATA Version is:       SATA 2.6, 6.0 Gb/s (current: 6.0 Gb/s)
Local Time is:         Fri Aug 28 19:20:55 2026 CEST
SMART support is:      Available - device has SMART capability.
SMART support is:      Enabled
AAM feature is:        Unavailable
APM feature is:        Unavailable
Rd look-ahead is:      Enabled
Write cache is:        Disabled
DSN feature is:        Unavailable
ATA Security is:       Unavailable
Wt Cache Reorder:      Enabled

=== START OF READ SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

General SMART Values:
Offline data collection status:  (0x02) Offline data collection activity
                                   was completed without error.
                                   Auto Offline Data Collection: Disabled.
Self-test execution status:      (   0) The previous self-test routine completed
                                   without error or no self-test has ever
                                   been run.
Total time to complete Offline
data collection:                 (   2) seconds.
Offline data collection
capabilities:                     (0x79) SMART execute Offline immediate.
                                   No Auto Offline data collection support.
                                   Suspend Offline collection upon new
                                   command.
                                   Offline surface scan supported.
                                   Self-test supported.
                                   Conveyance Self-test supported.
                                   Selective Self-test supported.
SMART capabilities:              (0x0003) Saves SMART data before entering
                                   power-saving mode.
                                   Supports SMART auto save timer.
Error logging capability:        (0x01) Error logging supported.
                                   General Purpose Logging supported.
Short self-test routine
recommended polling time:        (   1) minutes.
Extended self-test routine
recommended polling time:        (   2) minutes.
Conveyance self-test routine
recommended polling time:        (   2) minutes.
SCT capabilities:               (0x0039) SCT Status supported.
                                   SCT Error Recovery Control supported.
                                   SCT Feature Control supported.
                                   SCT Data Table supported.

SMART Attributes Data Structure revision number: 1
Vendor Specific SMART Attributes with Thresholds:
```

ID#	ATTRIBUTE_NAME	FLAGS	VALUE	WORST	THRESH	FAIL	RAW_VALUE
1	Raw_Read_Error_Rate	POSR--	130	130	039	-	18685971
5	Reallocated_Sector_Ct	PO--CK	097	097	001	-	0
9	Power_On_Hours	-O--CK	100	100	000	-	92509
12	Power_Cycle_Count	-O--CK	100	100	000	-	167
171	Unknown_Attribute	-O--CK	100	100	000	-	0
172	Unknown_Attribute	-O--CK	100	100	000	-	0
173	Unknown_Attribute	PO--CK	092	092	001	-	0
174	Unknown_Attribute	-O--CK	100	100	000	-	157
175	Program_Fail_Count_Chip	PO--CK	100	100	010	-	2302591238768
180	Unused_Rsvd_Blkc_Cnt_Tot	PO-RCK	130	130	039	-	74389253
184	End-to-End_Error	PO--CK	100	100	000	-	0
187	Reported_Uncorrect	-O--CK	100	100	000	-	0
188	Command_Timeout	-O--CK	100	100	000	-	0
190	Airflow_Temperature_Cel	-O---K	073	073	000	-	27 (Min/Max 20/27)
194	Temperature_Celsius	-O---K	100	100	000	-	27
195	Hardware_ECC_Recovered	-O-RC-	100	100	000	-	4652951
196	Reallocated_Event_Count	PO--CK	100	100	010	-	0
197	Current_Pending_Sector	-O--C-	100	100	000	-	0
198	Offline_Uncorrectable	---C-	100	100	000	-	0
199	UDMA_CRC_Error_Count	-OSRCK	100	100	000	-	0
226	Unknown_SSD_Attribute	-O--CK	100	100	000	-	31
227	Unknown_SSD_Attribute	-O--CK	100	100	000	-	25
228	Power-off_Retract_Count	-O--CK	100	100	000	-	7458

|||||_ K auto-keep
 |||||_ C event count
 |||||_ R error rate
 |||____ S speed/performance
 ||____ O updated online
 |____ P prefailure warning

General Purpose Log Directory Version 1

SMART Log Directory Version 1 [multi-sector log support]

Address	Access	R/W	Size	Description
0x00	GPL,SL	R/O	1	Log Directory
0x01	GPL,SL	R/O	1	Summary SMART error log
0x02	GPL,SL	R/O	8	Comprehensive SMART error log
0x03	GPL,SL	R/O	20	Ext. Comprehensive SMART error log
0x04	GPL,SL	R/O	8	Device Statistics log
0x06	GPL,SL	R/O	1	SMART self-test log
0x07	GPL,SL	R/O	2	Extended self-test log
0x09	GPL,SL	R/W	1	Selective self-test log
0x10	GPL,SL	R/O	1	NCQ Command Error log
0x11	GPL,SL	R/O	1	SATA Phy Event Counters log
0x80-0x9f	GPL,SL	R/W	16	Host vendor specific log
0xb0	GPL,SL	VS	1	Device vendor specific log
0xb1	GPL,SL	VS	33	Device vendor specific log
0xb6	GPL,SL	VS	127	Device vendor specific log
0xd0	GPL,SL	VS	1	Device vendor specific log
0xd1	GPL	VS	13056	Device vendor specific log
0xdf	GPL	VS	1152	Device vendor specific log
0xdf	SL	VS	128	Device vendor specific log
0xe0	GPL,SL	R/W	1	SCT Command/Status
0xe1	GPL,SL	R/W	1	SCT Data Transfer

SMART Extended Comprehensive Error Log Version: 1 (20 sectors)

Device Error Count: 4

CR = Command Register
 FEATR = Features Register
 COUNT = Count (was: Sector Count) Register
 LBA_48 = Upper bytes of LBA High/Mid/Low Registers] ATA-8
 LH = LBA High (was: Cylinder High) Register] LBA
 LM = LBA Mid (was: Cylinder Low) Register] Register
 LL = LBA Low (was: Sector Number) Register]
 DV = Device (was: Device/Head) Register
 DC = Device Control Register
 ER = Error register
 ST = Status register

Powered_Up_Time is measured from power on, and printed as

DDd+hh:mm:ss.sss where DD=days, hh=hours, mm=minutes,
SS=sec, and sss=millisec. It "wraps" after 49.710 days.

Error 4 [3] occurred at disk power-on lifetime: 5962 hours (248 days + 10 hours)
When the command that caused the error occurred, the device was active or idle.

After command completion occurred, registers were:

ER -- ST COUNT LBA_48 LH LM LL DV DC

```
-- -- == -- == == -- -- -- --
00 -- 51 02 00 00 5b ce 00 cc 00 40 00 Error: at LBA = 0x5bce00cc00 = 394298182656
```

Commands leading to the command that caused the error were:

CR	FEATR	COUNT	LBA_48	LH	LM	LL	DV	DC	Powered_Up_Time	Command/Feature_Name
60	00	00	02 00 00 5b ce 00 cc 00 40 00	11d+22:03:25.810	READ FPDMA QUEUED					
60	00	00	02 00 00 5b d0 00 cc 00 40 00	11d+22:03:25.806	READ FPDMA QUEUED					
60	00	00	02 00 00 5b d2 00 cc 00 40 00	11d+22:03:25.802	READ FPDMA QUEUED					
60	00	00	02 00 00 5b d4 00 cc 00 40 00	11d+22:03:25.798	READ FPDMA QUEUED					
60	00	00	02 00 00 5b d6 00 cc 00 40 00	11d+22:03:25.794	READ FPDMA QUEUED					

Error 3 [2] occurred at disk power-on lifetime: 2641 hours (110 days + 1 hours)
When the command that caused the error occurred, the device was active or idle.

After command completion occurred, registers were:

```
ER -- ST COUNT LBA_48 LH LM LL DV DC
-- -- == -- == == -- -- -- --
00 -- 51 02 00 00 04 72 00 a0 00 40 00 Error: at LBA = 0x47200a000 = 19092512768
```

Commands leading to the command that caused the error were:

CR	FEATR	COUNT	LBA_48	LH	LM	LL	DV	DC	Powered_Up_Time	Command/Feature_Name
60	00	00	02 00 00 04 72 00 a0 00 40 00	1d+00:06:06.919	READ FPDMA QUEUED					
60	00	00	02 00 00 04 70 00 a0 00 40 00	1d+00:06:06.913	READ FPDMA QUEUED					
60	00	00	02 00 00 04 6e 00 a0 00 40 00	1d+00:06:06.909	READ FPDMA QUEUED					
60	00	00	02 00 00 04 6c 00 a0 00 40 00	1d+00:06:06.906	READ FPDMA QUEUED					
60	00	00	02 00 00 04 6a 00 a0 00 40 00	1d+00:06:06.902	READ FPDMA QUEUED					

Error 2 [1] occurred at disk power-on lifetime: 2616 hours (109 days + 0 hours)
When the command that caused the error occurred, the device was active or idle.

After command completion occurred, registers were:

```
ER -- ST COUNT LBA_48 LH LM LL DV DC
-- -- == -- == == -- -- -- --
00 -- 51 02 00 00 06 2e 00 9f 00 40 00 Error: at LBA = 0x62e009f00 = 26541596416
```

Commands leading to the command that caused the error were:

CR	FEATR	COUNT	LBA_48	LH	LM	LL	DV	DC	Powered_Up_Time	Command/Feature_Name
60	00	00	02 00 00 06 2e 00 9f 00 40 00	00:23:49.793	READ FPDMA QUEUED					
60	00	00	02 00 00 06 2c 00 9f 00 40 00	00:23:49.787	READ FPDMA QUEUED					
60	00	00	02 00 00 06 2a 00 9f 00 40 00	00:23:49.784	READ FPDMA QUEUED					
60	00	00	02 00 00 06 28 00 9f 00 40 00	00:23:49.780	READ FPDMA QUEUED					
60	00	00	02 00 00 06 26 00 9f 00 40 00	00:23:49.776	READ FPDMA QUEUED					

Error 1 [0] occurred at disk power-on lifetime: 0 hours (0 days + 0 hours)
When the command that caused the error occurred, the device was active or idle.

After command completion occurred, registers were:

```
ER -- ST COUNT LBA_48 LH LM LL DV DC
-- -- == -- == == -- -- -- --
00 -- 51 02 00 00 0b 6c 00 e6 00 40 00 Error: at LBA = 0xb6c00e600 = 49056638464
```

Commands leading to the command that caused the error were:

CR	FEATR	COUNT	LBA_48	LH	LM	LL	DV	DC	Powered_Up_Time	Command/Feature_Name
60	00	00	02 00 00 0b 6c 00 e6 00 40 00	00:02:10.987	READ FPDMA QUEUED					
60	00	00	02 00 00 0b 6a 00 e6 00 40 00	00:02:10.986	READ FPDMA QUEUED					
60	00	00	02 00 00 0b 68 00 e6 00 40 00	00:02:10.984	READ FPDMA QUEUED					
60	00	00	02 00 00 0b 66 00 e6 00 40 00	00:02:10.983	READ FPDMA QUEUED					
60	00	00	02 00 00 0b 64 00 e6 00 40 00	00:02:10.981	READ FPDMA QUEUED					

SMART Extended Self-test Log Version: 1 (2 sectors)

Num	Test_Description	Status	Remaining	LifeTime(hours)	LBA_of_first_error
# 1	Short offline	Completed without error	00%	26953	-
# 2	Extended offline	Completed without error	00%	26768	-

SMART Selective self-test log data structure revision number 1

SPAN	MIN_LBA	MAX_LBA	CURRENT_TEST_STATUS
1	0	0	Not_testing
2	0	0	Not_testing
3	0	0	Not_testing
4	0	0	Not_testing
5	0	0	Not_testing

Selective self-test flags (0x0):

After scanning selected spans, do NOT read-scan remainder of disk.

If Selective self-test is pending on power-up, resume after 0 minute delay.

SCT Status Version: 3
SCT Version (vendor specific): 1 (0x0001)
Device State: Active (0)
Current Temperature: 27 Celsius
Power Cycle Min/Max Temperature: 20/27 Celsius
Lifetime Min/Max Temperature: 8/47 Celsius
Specified Max Operating Temperature: 55 Celsius
Under/Over Temperature Limit Count: 0/0
Vendor specific:
00 00 5f 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

SCT Temperature History Version: 2
Temperature Sampling Period: 1 minute
Temperature Logging Interval: 1 minute
Min/Max recommended Temperature: 0/55 Celsius
Min/Max Temperature Limit: 0/55 Celsius
Temperature History Size (Index): 478 (296)

Index	Estimated Time	Temperature Celsius
297	2026-08-28 11:23	25 *****
...	..(41 skipped).	.. *****
339	2026-08-28 12:05	25 *****
340	2026-08-28 12:06	26 *****
...	..(115 skipped).	.. *****
456	2026-08-28 14:02	26 *****
457	2026-08-28 14:03	25 *****
...	..(213 skipped).	.. *****
193	2026-08-28 17:37	25 *****
194	2026-08-28 17:38	26 *****
...	..(76 skipped).	.. *****
271	2026-08-28 18:55	26 *****
272	2026-08-28 18:56	27 *****
...	..(23 skipped).	.. *****
296	2026-08-28 19:20	27 *****

SCT Error Recovery Control:
Read: Disabled
Write: Disabled

Device Statistics (GP Log 0x04)

Page	Offset	Size	Value	Flags	Description
0x01	=====	=	=	===	== General Statistics (rev 2) ==
0x01	0x008	4	167	---	Lifetime Power-On Resets
0x01	0x018	6	80265037638	---	Logical Sectors Written
0x01	0x020	6	4011267	---	Number of Write Commands
0x01	0x028	6	1759452622605	---	Logical Sectors Read
0x01	0x030	6	148889	---	Number of Read Commands
0x04	=====	=	=	===	== General Errors Statistics (rev 1) ==
0x04	0x008	4	0	---	Number of Reported Uncorrectable Errors
0x04	0x010	4	0	---	Resets Between Cmd Acceptance and Completion
0x05	=====	=	=	===	== Temperature Statistics (rev 1) ==
0x05	0x008	1	27	---	Current Temperature
0x05	0x010	1	25	---	Average Short Term Temperature
0x05	0x018	1	18	---	Average Long Term Temperature
0x05	0x020	1	46	---	Highest Temperature
0x05	0x028	1	11	---	Lowest Temperature
0x05	0x030	1	31	---	Highest Average Short Term Temperature
0x05	0x038	1	11	---	Lowest Average Short Term Temperature
0x05	0x040	1	29	---	Highest Average Long Term Temperature
0x05	0x048	1	13	---	Lowest Average Long Term Temperature
0x05	0x050	4	0	---	Time in Over-Temperature
0x05	0x058	1	55	---	Specified Maximum Operating Temperature
0x05	0x060	4	0	---	Time in Under-Temperature
0x05	0x068	1	0	---	Specified Minimum Operating Temperature
0x06	=====	=	=	===	== Transport Statistics (rev 1) ==
0x06	0x008	4	78018	---	Number of Hardware Resets
0x06	0x010	4	487	---	Number of ASR Events
0x06	0x018	4	0	---	Number of Interface CRC Errors
0x07	=====	=	=	===	== Solid State Device Statistics (rev 1) ==
0x07	0x008	1	8	---	Percentage Used Endurance Indicator
				_	C monitored condition met
				__	D supports DSN
				___	N normalized value

Pending Defects log (GP Log 0x0c) not supported

SATA Phy Event Counters (GP Log 0x11)

ID	Size	Value	Description
0x0001	4	0	Command failed due to ICRC error
0x0003	4	0	R_ERR response for device-to-host data FIS
0x0004	4	0	R_ERR response for host-to-device data FIS
0x0006	4	0	R_ERR response for device-to-host non-data FIS
0x000a	4	27	Device-to-host register FISes sent due to a COMRESET
0x000b	4	0	CRC errors within host-to-device FIS
0x000d	4	0	Non-CRC errors within host-to-device FIS

===== Selbsttestprotokoll =====
smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.18.42-production+truenas] (local build)
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=== START OF READ SMART DATA SECTION ===

SMART Self-test log structure revision number 1

Num	Test_Description	Status	Remaining	LifeTime(hours)	LBA_of_first_error
# 1	Short offline	Completed without error	00%	26953	-
# 2	Extended offline	Completed without error	00%	26768	-

SMART-Bericht Supermicro X13SAE-F · erhoben am 27. und 28. August 2026

Sämtliche Werte stammen aus der unveränderten Ausgabe von smartctl auf dem laufenden System.

Auffälligkeiten sind benannt, nicht weggelassen.